MVP2 Mobile Apps Specification

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Helix VPN Mobile Client — Android, iOS, and HarmonyOS

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Related Documents: MVP2 Architecture Spec, MVP2 Rust Core Spec, MVP2 Desktop Spec

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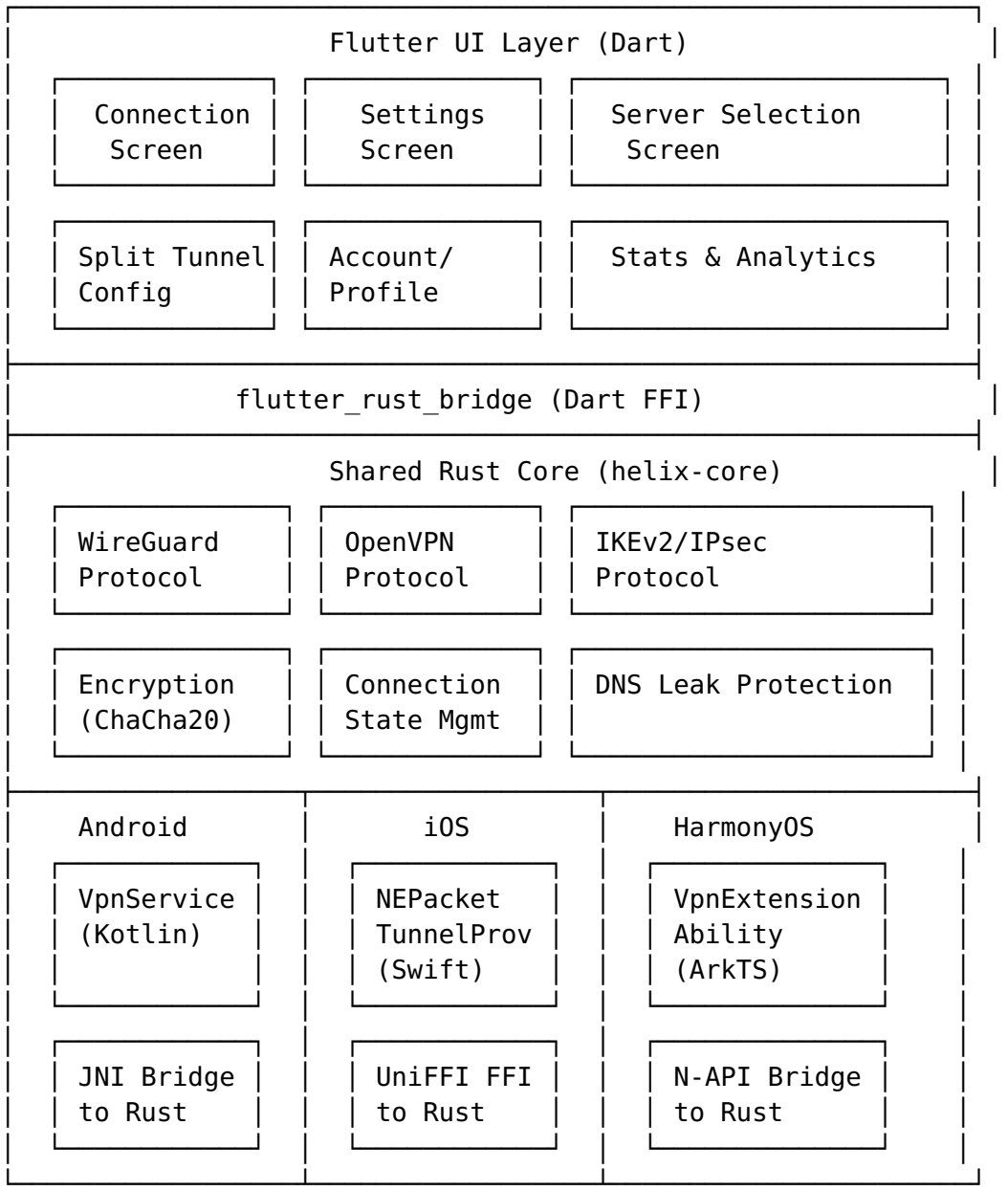
1. Mobile Client Overview

1.1 Unified Mobile Strategy

The Helix VPN mobile strategy employs a **unified Flutter codebase** with platform-specific native integrations for VPN tunnel management. This approach achieves ~85-90% code reuse across Android, iOS, and HarmonyOS while leveraging each platform's native VPN APIs for secure, performant tunneling.



1.2 Platform-Specific Adaptations

Aspect	Android	iOS	HarmonyOS
VPN API		`NEPacketTunnelProvider`	`VpnExtensionAbility`

Aspect	Android	iOS	HarmonyOS
	`VpnService` + `Builder`		
Native Language	Kotlin	Swift	ArkTS + C++ (N-API)
Bridge to Rust	JNI via flutter_rust_bridge	UniFFI + flutter_rust_bridge	N-API via flutter_rust_bridge
Background Model	Foreground Service	System-managed extension	ExtensionAbility lifecycle
Credential Store	Android Keystore	iOS Keychain	HUKS (Huawei Universal Keystore)
Push Notifications	FCM (Firebase)	APNs	HMS Push Kit
Min OS Version	Android 8.0 (API 26)	iOS 15.0	HarmonyOS NEXT (API 12)

1.3 System Requirements

Android:

- Minimum SDK: API 26 (Android 8.0 Oreo)
- Target SDK: API 35 (Android 15)
- Architecture: arm64-v8a, armeabi-v7a, x86_64 (for emulator)
- Permissions: `BIND_VPN_SERVICE`, `FOREGROUND_SERVICE`, `FOREGROUND_SERVICE_SPECIAL_USE`, `FOREGROUND_SERVICE_DATA_SYNC`, `POST_NOTIFICATIONS`, `ACCESS_NETWORK_STATE`, `INTERNET`

iOS:

- Minimum Deployment Target: iOS 15.0
- Architectures: arm64 (devices), arm64-sim (simulator where supported)
- Entitlements: `com.apple.developer.networking.networkextension` (packet-tunnel-provider)
- Capabilities: Network Extensions, App Groups, Keychain Sharing
- Physical device required for VPN testing (simulator does not support Network Extensions)

HarmonyOS:

- Minimum API: 12 (HarmonyOS NEXT)
- Architecture: arm64-v8a
- Permissions: `ohos.permission.MANAGE_VPN`, `ohos.permission.INTERNET`, `ohos.permission.GET_NETWORK_INFO`
- ACL required for `MANAGE_VPN` permission (system_grant)

1.4 Architecture Principles

1. **Rust-First Core:** All VPN protocol logic, encryption, and state management lives in the shared `helix-core` Rust library. Native platform code is a thin adapter layer.
2. **Memory-Conscious iOS:** The iOS NEPacketTunnelProvider operates under a strict ~15MB memory limit. The Swift wrapper must be minimal; all heavy lifting is done in optimized Rust code.
3. **Foreground Service for Android:** Android requires a persistent foreground service with notification. The service must handle Doze mode, manufacturer-specific battery optimizations, and Android 15's 6-hour foreground service limit.
4. **UniFFI for Binding Generation:** Cross-platform Rust bindings for Kotlin and Swift are auto-generated using Mozilla's UniFFI, reducing hand-written binding code and potential memory safety issues.

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2. Android Client

2.1 Architecture

The Android client uses a three-layer architecture:

```
┌─────────────────────────────────────────────┐
│         Flutter UI Layer (Dart)             │
│  - Connection screen, settings, stats       │
│  - MethodChannel for native calls           │
├─────────────────────────────────────────────┤
│      flutter_rust_bridge Layer              │
│  - Auto-generated Dart ↔ Rust bindings      │
│  - Handles memory safety at FFI boundary    │
├─────────────────────────────────────────────┤
│      Native Platform Layer (Kotlin)         │
│  - HelixVpnService extends VpnService       │
│  - MethodChannel handler for Flutter        │
│  - Foreground service management            │
│  - Notification builder                     │
│  - Quick Settings TileService               │
├─────────────────────────────────────────────┤
│         Shared Rust Core                    │
│  - helix-core via JNI                       │
│  - WireGuard/OpenVPN protocol impl          │
│  - Encryption & packet routing              │
└─────────────────────────────────────────────┘
```

2.2 VpnService Implementation

2.2.1 Builder Pattern for Interface Configuration

The `HelixVpnService` extends Android's `VpnService` and uses the `Builder` inner class to configure the virtual network interface:

```
class HelixVpnService : VpnService() {

    companion object {
        const val ACTION_CONNECT = "com.helix.vpn.CONNECT"
        const val ACTION_DISCONNECT = "com.helix.vpn.DISCONNECT"
        const val NOTIFICATION_ID = 1
        const val CHANNEL_ID = "helix_vpn_channel"

        @JvmStatic
        var isRunning = false
            private set

        @JvmStatic
        var connectionState: VpnConnectionState =
        VpnConnectionState.DISCONNECTED
            private set
    }

    private var vpnInterface: ParcelFileDescriptor? = null
    private var tunnelThread: Thread? = null
    private val rustBridge = HelixRustBridge() // JNI wrapper for
        helix-core

    override fun onStartCommand(intent: Intent?, flags: Int,
        startId: Int): Int {
        when (intent?.action) {
            ACTION_CONNECT -> {
                val config =
        intent.getParcelableExtra<VpnConfig>("config")
                    ?: return START_NOT_STICKY
                startVpnConnection(config)
            }
            ACTION_DISCONNECT -> {
                stopVpnConnection()
            }
            else -> {
                // Service restarted by system (always-on VPN)
                val savedConfig = loadSavedConfiguration()
                if (savedConfig != null) {
                    startVpnConnection(savedConfig)
                }
            }
        }
        return START_STICKY // Restart if killed by system
    }
```

```
private fun startVpnConnection(config: VpnConfig) {
    // Build VPN interface configuration
    val builder = Builder().apply {
        setSession("Helix VPN")
        setMtu(config.mtu) // Typically 1420 for WireGuard

        // Configure tunnel IP address
        addAddress(config.tunnelAddress,
    config.tunnelPrefixLength)

        // Configure DNS servers (critical for leak
    prevention)
        config.dnsServers.forEach { dns ->
            addDnsServer(dns)
        }

        // Add search domains
        config.searchDomains.forEach { domain ->
            addSearchDomain(domain)
        }

        // Route configuration
        if (config.routeAllTraffic) {
            addRoute("0.0.0.0", 0)  // IPv4 default route
            addRoute("::", 0)        // IPv6 default route
        } else {
            config.includedRoutes.forEach { route ->
                addRoute(route.address, route.prefixLength)
            }
        }

        // Exclude routes (Android 13+ / API 33+)
        if (Build.VERSION.SDK_INT >=
    Build.VERSION_CODES.TIRAMISU) {
            config.excludedRoutes.forEach { route ->

    excludeRoute(IpPrefix(InetAddress.getByName(route.address),
    route.prefixLength))
            }
        }

        // Split tunneling: allowed applications (whitelist
    mode)
        if (config.allowedApplications.isNotEmpty()) {
            config.allowedApplications.forEach { packageName
    ->
                try {
                    addAllowedApplication(packageName)
                } catch (e:
    PackageManager.NameNotFoundException) {
                    Log.w("HelixVPN", "Package not found:
    $packageName")
```

```
            }
        }
    }

    // Split tunneling: disallowed applications (blacklist
mode)
    // Mutually exclusive with allowedApplications
    if (config.disallowedApplications.isNotEmpty() &&
config.allowedApplications.isEmpty()) {
        config.disallowedApplications.forEach {
packageName ->
            try {
                addDisallowedApplication(packageName)
            } catch (e:
PackageManager.NameNotFoundException) {
                Log.w("HelixVPN", "Package not found:
$packageName")
            }
        }
    }

    // Allow apps to bypass VPN using bindProcessToNetwork
    if (config.allowBypass) {
        allowBypass()
    }

    // HTTP proxy configuration (Android 10+ / API 29+)
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.Q &&
config.httpProxy != null) {
        val proxyInfo = ProxyInfo.buildDirectProxy(
            config.httpProxy.host,
            config.httpProxy.port
        )
        setHttpProxy(proxyInfo)
    }

    // Blocking mode for the TUN file descriptor
    setBlocking(true)
}

// Establish the VPN interface (creates TUN device)
vpnInterface = builder.establish()

if (vpnInterface != null) {
    // Start foreground service with persistent
notification
    val notification =
buildVpnNotification(config.serverLocation)
    if (Build.VERSION.SDK_INT >=
Build.VERSION_CODES.UPSIDE_DOWN_CAKE) {
        startForeground(NOTIFICATION_ID, notification,

ServiceInfo.FOREGROUND_SERVICE_TYPE_SPECIAL_USE)
```

```
            } else {
                startForeground(NOTIFICATION_ID, notification)
            }

            // Pass TUN fd to Rust core for packet processing
            val tunFd = vpnInterface!!.fd
            rustBridge.startTunnel(tunFd, config.toRustConfig())

            isRunning = true
            connectionState = VpnConnectionState.CONNECTED
            broadcastStateChange(connectionState)
        }
    }

    private fun stopVpnConnection() {
        rustBridge.stopTunnel()
        vpnInterface?.close()
        vpnInterface = null
        isRunning = false
        connectionState = VpnConnectionState.DISCONNECTED
        broadcastStateChange(connectionState)
        stopForeground(STOP_FOREGROUND_REMOVE)
        stopSelf()
    }

    override fun onRevoke() {
        // Called when user revokes VPN permission or another VPN
        app takes over
        Log.w("HelixVPN", "VPN permission revoked by system or
        user")
        stopVpnConnection()
        broadcastStateChange(VpnConnectionState.REVOKED)
    }

    override fun onDestroy() {
        stopVpnConnection()
        super.onDestroy()
    }
}
```

2.2.2 Split Tunneling Configuration

Android provides the most comprehensive split tunneling on mobile through two mutually exclusive mechanisms:

Whitelist Mode (allowedApplications):

```
// Only specified apps use the VPN; all other traffic bypasses
builder.addAllowedApplication("com.example.corporate_app")
builder.addAllowedApplication("com.example.secure_browser")
// All other apps connect directly
```

Blacklist Mode (disallowedApplications):

```
// Specified apps bypass the VPN; all other traffic goes through
        VPN
builder.addDisallowedApplication("com.bank.app")        // Banking
        bypasses VPN
builder.addDisallowedApplication("com.netflix.app")     //
        Streaming bypasses VPN
builder.addDisallowedApplication("com.spotify.music")   // Music
        bypasses VPN
```

IP-based Split Tunneling (API 33+):

```
if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
    // Exclude LAN traffic from VPN
    builder.excludeRoute(IpPrefix(InetAddress.getByName("192.168.0.0"), 16))
    builder.excludeRoute(IpPrefix(InetAddress.getByName("10.0.0.0"), 8))
    builder.excludeRoute(IpPrefix(InetAddress.getByName("172.16.0.0"), 12))
}
```

Pre-API 33 Workaround: For devices running Android 12 and below, the app calculates all non-excluded routes using the WireGuard AllowedIPs calculator approach and adds them individually via `builder.addRoute()`.

2.2.3 Always-On VPN and Lockdown Mode

```
object AlwaysOnVpnHelper {

    /**
     * Check if this VPN is running as an always-on VPN.
     * Returns true if the system auto-started this service.
     */
    fun isAlwaysOn(context: Context): Boolean {
        return if (Build.VERSION.SDK_INT >=
        Build.VERSION_CODES.Q) {
            val vpnService = context as? VpnService
            vpnService?.isAlwaysOn ?: false
        } else {
            false
        }
    }

    /**
     * Check if lockdown mode (kill switch) is active.
     * When lockdown is enabled, all traffic is blocked if VPN
        disconnects.
     */
    fun isLockdownEnabled(context: Context): Boolean {
        return if (Build.VERSION.SDK_INT >=
        Build.VERSION_CODES.Q) {
            val vpnService = context as? VpnService
            vpnService?.isLockdownEnabled ?: false
```

```
        } else {
            false
        }
    }

    /**
     * Guide the user to system settings to enable always-on VPN.
     * Apps cannot programmatically enable always-on (requires
        system/MDM).
     */
    fun openVpnSettings(context: Context) {
        val intent = Intent(Settings.ACTION_VPN_SETTINGS)
        context.startActivity(intent)
    }

    /**
     * Metadata in AndroidManifest.xml to declare always-on
        support:
     * <meta-data
        android:name="android.net.VpnService.SUPPORTS_ALWAYS_ON"
     *            android:value="true" />
     */
}
```

AndroidManifest.xml declarations:

```
<service
    android:name=".vpn.HelixVpnService"
    android:permission="android.permission.BIND_VPN_SERVICE"
    android:foregroundServiceType="specialUse|dataSync"
    android:exported="true">
    <intent-filter>
        <action android:name="android.net.VpnService" />
    </intent-filter>
    <meta-data
        android:name="android.net.VpnService.SUPPORTS_ALWAYS_ON"
        android:value="true" />
    <property
        android:name="android.app.PROPERTY_SPECIAL_USE_FGS_SUBTYPE"
        android:value="vpn" />
</service>
```

2.3 Foreground Service with Persistent Notification

```
class VpnNotificationHelper(private val context: Context) {

    init {
        createNotificationChannel()
    }

    private fun createNotificationChannel() {
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
```

```
            val channel = NotificationChannel(
                HelixVpnService.CHANNEL_ID,
                "Helix VPN Connection",
                NotificationManager.IMPORTANCE_LOW
            ).apply {
                description = "Shows when Helix VPN is active"
                setShowBadge(false)
                enableLights(false)
                enableVibration(false)
                lockscreenVisibility =
        Notification.VISIBILITY_PUBLIC
            }

            val notificationManager =
        context.getSystemService(NotificationManager::class.java)
            notificationManager.createNotificationChannel(channel)
        }
    }

    fun buildVpnNotification(serverLocation: String, protocol:
        String = "WireGuard"): Notification {
        // Intent to open the app when notification is tapped
        val contentIntent = PendingIntent.getActivity(
            context,
            0,

        context.packageManager.getLaunchIntentForPackage(context.packageName),
            PendingIntent.FLAG_UPDATE_CURRENT or
        PendingIntent.FLAG_IMMUTABLE
        )

        // Intent to disconnect VPN
        val disconnectIntent = PendingIntent.getService(
            context,
            1,
            Intent(context, HelixVpnService::class.java).apply {
                action = HelixVpnService.ACTION_DISCONNECT
            },
            PendingIntent.FLAG_UPDATE_CURRENT or
        PendingIntent.FLAG_IMMUTABLE
        )

        // Intent to pause VPN (temporary disconnect)
        val pauseIntent = PendingIntent.getService(
            context,
            2,
            Intent(context, HelixVpnService::class.java).apply {
                action = "com.helix.vpn.PAUSE"
            },
            PendingIntent.FLAG_UPDATE_CURRENT or
        PendingIntent.FLAG_IMMUTABLE
        )
```

```
            return NotificationCompat.Builder(context,
            HelixVpnService.CHANNEL_ID)
                .setContentTitle("Helix VPN Connected")
                .setContentText("Server: $serverLocation \u00B7
            Protocol: $protocol")
                .setSmallIcon(R.drawable.ic_notification_vpn)
                .setContentIntent(contentIntent)
                .setOngoing(true) // Cannot be dismissed by user
                .setPriority(NotificationCompat.PRIORITY_LOW)
                .setCategory(NotificationCompat.CATEGORY_SERVICE)
                .addAction(R.drawable.ic_pause, "Pause", pauseIntent)
                .addAction(R.drawable.ic_disconnect, "Disconnect",
            disconnectIntent)
                .setForegroundServiceBehavior(NotificationCompat.FOREGROUND_SERVICE_IM
                .build()
        }

        fun updateNotification(serverLocation: String, stats:
            ConnectionStats) {
            val notification = buildVpnNotification(serverLocation)
            val notificationManager =
            context.getSystemService(NotificationManager::class.java)
            notificationManager.notify(HelixVpnService.NOTIFICATION_ID,
            notification)
        }
    }
```

2.4 Battery Optimization Handling

```
    object BatteryOptimizationHelper {

        /**
         * Check if the app is whitelisted from battery optimizations.
         * Foreground services should still function without this, but
         * some manufacturers aggressively kill non-whitelisted apps.
         */
        fun isIgnoringBatteryOptimizations(context: Context): Boolean
            {
            val powerManager =
            context.getSystemService(Context.POWER_SERVICE) as
            PowerManager
            return
            powerManager.isIgnoringBatteryOptimizations(context.packageName)
        }

        /**
         * Request battery optimization exemption.
         * NOTE: Google Play restricts use of
            ACTION_REQUEST_IGNORE_BATTERY_OPTIMIZATIONS.
         * Play Store apps should guide users to settings instead.
         */
        fun requestBatteryOptimizationExemption(context: Context) {
            // Safe approach for Google Play Store apps:
```

```
    val intent =
    Intent(Settings.ACTION_IGNORE_BATTERY_OPTIMIZATION_SETTINGS)
    context.startActivity(intent)
}

/**
 * For non-Play Store builds (F-Droid, direct APK):
 * Can directly request exemption.
 */
fun directRequestExemption(context: Context) {
    if (!isIgnoringBatteryOptimizations(context)) {
        val intent =
    Intent(Settings.ACTION_REQUEST_IGNORE_BATTERY_OPTIMIZATIONS).apply
    {
            data = Uri.parse("package:${context.packageName}")
        }
        context.startActivity(intent)
    }
}

/**
 * Manufacturer-specific battery optimization guides.
 * OEMs like Samsung, Xiaomi, OnePlus have additional
    restrictions.
 */
fun getManufacturerGuide(): String? {
    return when (Build.MANUFACTURER.lowercase()) {
        "samsung" -> "samsung_battery_guide"
        "xiaomi", "redmi", "poco" -> "xiaomi_battery_guide"
        "oneplus", "oppo" -> "oppo_battery_guide"
        "huawei", "honor" -> "huawei_battery_guide"
        "vivo", "iqoo" -> "vivo_battery_guide"
        else -> null
    }
}

/**
 * Monitor Doze mode changes to adjust keepalive intervals.
 */
fun registerDozeReceiver(context: Context, callback:
    (Boolean) -> Unit) {
    val receiver = object : BroadcastReceiver() {
        override fun onReceive(context: Context, intent:
    Intent) {
            val powerManager =
    context.getSystemService(Context.POWER_SERVICE) as
    PowerManager
            callback(powerManager.isDeviceIdleMode)
        }
    }
    context.registerReceiver(receiver,
    IntentFilter(PowerManager.ACTION_DEVICE_IDLE_MODE_CHANGED))
```

```
    }
}
```

2.5 Quick Settings Tile Implementation

```
class HelixVpnTileService : TileService() {

    private val serviceConnection = object : ServiceConnection {
        override fun onServiceConnected(name: ComponentName?,
        service: IBinder?) {}
        override fun onServiceDisconnected(name: ComponentName?)
        {}
    }

    override fun onStartListening() {
        super.onStartListening()
        updateTileState()
    }

    override fun onClick() {
        // Collapse the Quick Settings panel
        if (Build.VERSION.SDK_INT >=
        Build.VERSION_CODES.UPSIDE_DOWN_CAKE) {
            // Android 14+ requires different approach
        }

        when (qsTile.state) {
            Tile.STATE_INACTIVE -> {
                // Connect to VPN
                val intent = Intent(this,
        HelixVpnService::class.java).apply {
                    action = HelixVpnService.ACTION_CONNECT
                    putExtra("config", getQuickConnectConfig())
                }
                startService(intent)
            }
            Tile.STATE_ACTIVE -> {
                // Disconnect VPN
                val intent = Intent(this,
        HelixVpnService::class.java).apply {
                    action = HelixVpnService.ACTION_DISCONNECT
                }
                startService(intent)
            }
        }
    }

    private fun updateTileState() {
        val tile = qsTile ?: return

        when (HelixVpnService.connectionState) {
            VpnConnectionState.CONNECTED -> {
```

```
                tile.state = Tile.STATE_ACTIVE
                tile.label = "Helix VPN: ON"
                tile.icon = Icon.createWithResource(this,
        R.drawable.ic_tile_vpn_on)
            }
            VpnConnectionState.CONNECTING -> {
                tile.state = Tile.STATE_UNAVAILABLE
                tile.label = "Connecting..."
                tile.icon = Icon.createWithResource(this,
        R.drawable.ic_tile_vpn_connecting)
            }
            else -> {
                tile.state = Tile.STATE_INACTIVE
                tile.label = "Helix VPN"
                tile.icon = Icon.createWithResource(this,
        R.drawable.ic_tile_vpn_off)
            }
        }
        tile.updateTile()
    }

    private fun getQuickConnectConfig(): VpnConfig {
        // Load user's preferred quick-connect server
        val prefs = getSharedPreferences("helix_prefs",
        Context.MODE_PRIVATE)
        val serverId = prefs.getString("quick_connect_server",
        "auto") ?: "auto"
        return VpnConfig.quickConnect(serverId)
    }
}
```

AndroidManifest.xml registration:

```
<service
    android:name=".vpn.HelixVpnTileService"
    android:permission="android.permission.BIND_QUICK_SETTINGS_TILE"
    android:exported="true">
    <intent-filter>
        <action
        android:name="android.service.quicksettings.action.QS_TILE" /
        >
    </intent-filter>
    <meta-data
        android:name="android.service.quicksettings.ACTIVE_TILE"
        android:value="true" />
</service>
```

2.6 Kotlin Platform Channel Code

```
class MainActivity : FlutterActivity() {

    private val VPN_CHANNEL = "com.helix.vpn/native"
```

```
private lateinit var vpnMethodChannel: MethodChannel

override fun configureFlutterEngine(flutterEngine:
    FlutterEngine) {
    super.configureFlutterEngine(flutterEngine)

    vpnMethodChannel =
    MethodChannel(flutterEngine.dartExecutor.binaryMessenger,
    VPN_CHANNEL)
    vpnMethodChannel.setMethodCallHandler { call, result ->
        when (call.method) {
            "connect" -> handleConnect(call, result)
            "disconnect" -> handleDisconnect(result)
            "getStatus" -> handleGetStatus(result)
            "getStats" -> handleGetStats(result)
            "prepareVpn" -> handlePrepareVpn(result)
            "isVpnPrepared" -> handleIsVpnPrepared(result)
            "getInstalledApps" ->
    handleGetInstalledApps(result)
            "setSplitTunneling" ->
    handleSetSplitTunneling(call, result)
            "requestBatteryOptimizationExemption" ->
    handleBatteryExemption(result)
            else -> result.notImplemented()
        }
    }
}

private fun handleConnect(call: MethodCall, result:
    MethodChannel.Result) {
    val configMap = call.argument<Map<String, Any>>("config")
    val config = VpnConfig.fromMap(configMap ?: emptyMap())

    // Check VPN permission
    val intent = VpnService.prepare(this)
    if (intent != null) {
        // User hasn't granted VPN permission yet
        result.error("VPN_PERMISSION_REQUIRED", "User must
    grant VPN permission", null)
        return
    }

    // Start VPN service
    val serviceIntent = Intent(this,
    HelixVpnService::class.java).apply {
        action = HelixVpnService.ACTION_CONNECT
        putExtra("config", config)
    }

    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
        startForegroundService(serviceIntent)
    } else {
        startService(serviceIntent)
```

```
        }

        result.success(true)
    }

    private fun handleDisconnect(result: MethodChannel.Result) {
        val intent = Intent(this,
        HelixVpnService::class.java).apply {
            action = HelixVpnService.ACTION_DISCONNECT
        }
        startService(intent)
        result.success(true)
    }

    private fun handleGetStatus(result: MethodChannel.Result) {
        val status = when (HelixVpnService.connectionState) {
            VpnConnectionState.CONNECTED -> "connected"
            VpnConnectionState.CONNECTING -> "connecting"
            VpnConnectionState.DISCONNECTING -> "disconnecting"
            VpnConnectionState.DISCONNECTED -> "disconnected"
            VpnConnectionState.REVOKED -> "revoked"
            VpnConnectionState.ERROR -> "error"
        }
        result.success(status)
    }

    private fun handleGetStats(result: MethodChannel.Result) {
        val stats = rustBridge.getConnectionStats()
        result.success(mapOf(
            "uploadSpeed" to stats.uploadSpeed,
            "downloadSpeed" to stats.downloadSpeed,
            "totalUpload" to stats.totalUpload,
            "totalDownload" to stats.totalDownload,
            "duration" to stats.connectionDuration,
            "serverLocation" to stats.serverLocation,
            "protocol" to stats.protocol
        ))
    }

    private fun handlePrepareVpn(result: MethodChannel.Result) {
        val intent = VpnService.prepare(this)
        if (intent != null) {
            // Launch system VPN permission dialog
            vpnPermissionLauncher.launch(intent)
            result.success(false) // Permission not yet granted
        } else {
            result.success(true) // Already granted
        }
    }

    private fun handleIsVpnPrepared(result: MethodChannel.Result)
        {
```

```
        result.success(VpnService.prepare(this) == null)
    }

    private fun handleGetInstalledApps(result:
        MethodChannel.Result) {
        // Return list of installed apps for split tunneling
        configuration
        val pm = packageManager
        val apps =
        pm.getInstalledApplications(PackageManager.GET_META_DATA)
            .filter { app ->
                // Filter out system apps without launch intent
                app.flags and ApplicationInfo.FLAG_SYSTEM == 0 ||
                pm.getLaunchIntentForPackage(app.packageName) !=
        null
            }
            .map { app ->
                mapOf(
                    "packageName" to app.packageName,
                    "name" to
        pm.getApplicationLabel(app).toString(),
                    "icon" to app.packageName // Flutter side
        loads icon via package_name
                )
            }
            .sortedBy { it["name"] as String }

        result.success(apps)
    }

    private fun handleSetSplitTunneling(call: MethodCall, result:
        MethodChannel.Result) {
        val mode = call.argument<String>("mode") ?: "off"
        val apps = call.argument<List<String>>("apps") ?:
        emptyList()

        // Save split tunneling preferences
        val prefs = getSharedPreferences("helix_prefs",
        Context.MODE_PRIVATE)
        prefs.edit()
            .putString("split_tunnel_mode", mode)
            .putStringSet("split_tunnel_apps", apps.toSet())
            .apply()

        result.success(true)
    }

    private fun handleBatteryExemption(result:
        MethodChannel.Result) {
        BatteryOptimizationHelper.requestBatteryOptimizationExemption(this)
        result.success(true)
    }
```

```
    private val vpnPermissionLauncher = registerForActivityResult(
        ActivityResultContracts.StartActivityForResult()
    ) { result ->
        if (result.resultCode == RESULT_OK) {

        vpnMethodChannel.invokeMethod("onVpnPermissionGranted",
        null)
        } else {

        vpnMethodChannel.invokeMethod("onVpnPermissionDenied",
        null)
        }
    }
}
```

2.7 Gradle Build Configuration

```
// android/app/build.gradle
plugins {
    id "com.android.application"
    id "kotlin-android"
    id "dev.flutter.flutter-gradle-plugin"
    id "com.google.gms.google-services" // FCM (optional)
}

android {
    namespace "com.helix.vpn"
    compileSdk 35
    ndkVersion "25.2.9519653"

    compileOptions {
        sourceCompatibility JavaVersion.VERSION_17
        targetCompatibility JavaVersion.VERSION_17
    }

    kotlinOptions {
        jvmTarget = '17'
    }

    defaultConfig {
        applicationId "com.helix.vpn"
        minSdk 26
        targetSdk 35
        versionCode 100
        versionName "1.0.0"

        multiDexEnabled true

        ndk {
            abiFilters 'arm64-v8a', 'armeabi-v7a', 'x86_64'
        }
```

```
        externalNativeBuild {
            cmake {
                arguments "-DANDROID_STL=c++_shared"
            }
        }
    }

    // Product flavors for different distribution channels
    flavorDimensions += "distribution"

    productFlavors {
        playStore {
            dimension "distribution"
            applicationIdSuffix ".play"
            buildConfigField "boolean",
        "ENABLE_DIRECT_BATTERY_REQUEST", "false"
        }
        fdroid {
            dimension "distribution"
            applicationIdSuffix ".fdroid"
            buildConfigField "boolean",
        "ENABLE_DIRECT_BATTERY_REQUEST", "true"
        }
        direct {
            dimension "distribution"
            buildConfigField "boolean",
        "ENABLE_DIRECT_BATTERY_REQUEST", "true"
        }
    }

    buildTypes {
        release {
            minifyEnabled true
            shrinkResources true
            proguardFiles getDefaultProguardFile('proguard-
        android-optimize.txt'), 'proguard-rules.pro'
            signingConfig signingConfigs.release
        }
        debug {
            minifyEnabled false
            signingConfig signingConfigs.debug
        }
    }

    externalNativeBuild {
        cmake {
            path "src/main/cpp/CMakeLists.txt"
            version "3.22.1"
        }
    }
}
```

```
flutter {
    source '../..'
}

dependencies {
    implementation "org.jetbrains.kotlin:kotlin-stdlib-jdk8"
    implementation 'androidx.core:core-ktx:1.13.0'
    implementation 'androidx.appcompat:appcompat:1.7.0'

    // Rust JNI bridge
    implementation project(':rust_bridge')

    // FCM for push notifications (optional)
    playStoreImplementation 'com.google.firebase:firebase-
        messaging:24.0.0'
}
```

2.8 ProGuard / R8 Rules

```
# android/app/proguard-rules.pro

# Keep VPN service and related classes
-keep class com.helix.vpn.vpn.** { *; }
-keep class com.helix.vpn.model.** { *; }

# Keep Rust JNI bridge classes
-keep class com.helix.vpn.rust.** { *; }
-keepclasseswithmembernames class * {
    native <methods>;
}

# Keep Flutter plugin classes
-keep class io.flutter.plugins.** { *; }
-keep class io.flutter.embedding.** { *; }

# Keep Parcelable implementations
-keepclassmembers class * implements android.os.Parcelable {
    public static final ** CREATOR;
}

# Keep serialized models for MethodChannel
-keepclassmembers class com.helix.vpn.model.VpnConfig { *; }
-keepclassmembers class com.helix.vpn.model.ConnectionStats { *; }

# Rust library
-keep class com.helix.vpn.rust.HelixRustBridge { *; }

# Don't warn about missing platform classes
-dontwarn java.awt.**
-dontwarn javax.swing.**
```

2.9 App Signing and Google Play Distribution

```
// android/app/build.gradle (signing configuration)
android {
    signingConfigs {
        release {
            storeFile
        file(System.getenv("HELIX_KEYSTORE_PATH") ?: "helix-
        release.keystore")
            storePassword System.getenv("HELIX_KEYSTORE_PASSWORD")
            keyAlias System.getenv("HELIX_KEY_ALIAS")
            keyPassword System.getenv("HELIX_KEY_PASSWORD")
        }
    }
}
```

AAB (Android App Bundle) for Google Play:

```
# Build release AAB
flutter build appbundle --release --flavor playStore

# Output: build/app/outputs/bundle/playStoreRelease/app-playStore-
        release.aab
```

APK for direct distribution / F-Droid:

```
# Build split APKs per ABI
flutter build apk --release --split-per-abi --flavor fdroid

# Output:
# build/app/outputs/apk/fdroid/release/app-arm64-v8a-fdroid-
        release.apk
# build/app/outputs/apk/fdroid/release/app-armeabi-v7a-fdroid-
        release.apk
# build/app/outputs/apk/fdroid/release/app-x86_64-fdroid-
        release.apk
```

2.10 Android 15+ Foreground Service Restrictions

Starting with Android 15 (API 35), all foreground services share a **6-hour time limit** within a 24-hour window. For a VPN app, this requires adaptation:

```
object Android15ForegroundServiceHelper {

    /**
     * For Android 15+, use User-Initiated Data Transfer (UIDT)
        Jobs
     * as a fallback when foreground service time limit is
        reached.
     */
    @RequiresApi(Build.VERSION_CODES.UPSIDE_DOWN_CAKE)
    fun scheduleVpnKeepAliveJob(context: Context) {
```

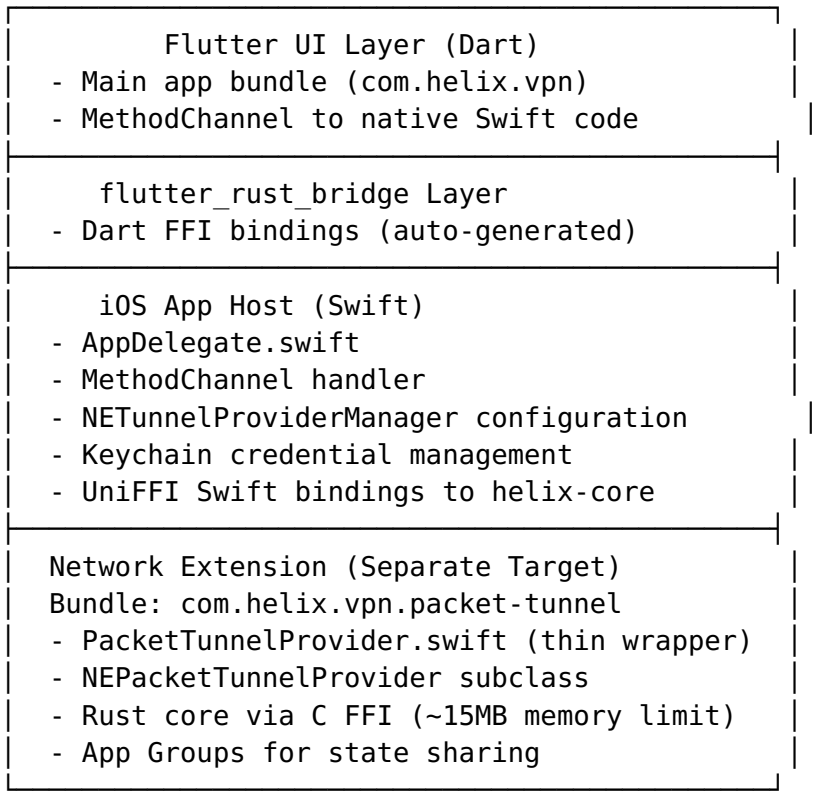
```
        val jobScheduler =
        context.getSystemService(Context.JOB_SCHEDULER_SERVICE)
        as JobScheduler

        val jobInfo = JobInfo.Builder(
            JOB_ID_VPN_KEEPALIVE,
            ComponentName(context,
        VpnKeepAliveJobService::class.java)
        ).apply {
            setUserInitiated(true)
            setPersisted(true) // Survive reboots
            setRequiredNetworkType(JobInfo.NETWORK_TYPE_ANY)
            setMinimumLatency(TimeUnit.MINUTES.toMillis(5))
        }.build()

        jobScheduler.schedule(jobInfo)
    }

    /**
     * UIDT JobService that runs when the foreground service needs
        a break.
     * Maintains the VPN tunnel by keeping the Rust core active.
     */
    class VpnKeepAliveJobService : JobService() {
        override fun onStartJob(params: JobParameters?): Boolean {
            if (HelixVpnService.isRunning) {
                // Ensure VPN is still active; restart foreground
        service if needed
                val serviceIntent = Intent(this,
        HelixVpnService::class.java)
                startForegroundService(serviceIntent)
            }
            jobFinished(params, false)
            return true
        }

        override fun onStopJob(params: JobParameters?): Boolean {
            return true // Reschedule if stopped
        }
    }

    private const val JOB_ID_VPN_KEEPALIVE = 1001
}
```

---

3. iOS Client

3.1 Architecture

The iOS client uses a four-layer architecture with a separate Network Extension target:



3.2 NEPacketTunnelProvider Implementation

3.2.1 PacketTunnelProvider Extension

```swift
// PacketTunnelProvider/PacketTunnelProvider.swift
import NetworkExtension
import os.log

// Rust FFI imports (generated by UniFFI or hand-written)
import helix_core // Rust library C FFI bindings

class PacketTunnelProvider: NEPacketTunnelProvider {

    private let logger = OSLog(subsystem: "com.helix.vpn",
        category: "PacketTunnel")
    private var rustTunnelHandle:
        OpaquePointer? // Handle to Rust tunnel instance
    private var isTunnelActive = false

    // MARK: - Tunnel Lifecycle
```

```
override func startTunnel(
    options: [String: NSObject]?,
    completionHandler: @escaping (Error?) -> Void
) {
    os_log("Starting tunnel...", log: logger, type: .info)

    // Extract configuration from protocolConfiguration
    guard let proto = protocolConfiguration as?
    NETunnelProviderProtocol,
          let providerConfig = proto.providerConfiguration
    else {
        completionHandler(NEVPNError(.configurationInvalid))
        return
    }

    // Retrieve configuration from App Group shared storage
    let config = loadConfiguration(from: providerConfig)

    // Configure tunnel network settings
    let settings = buildTunnelNetworkSettings(config: config)

    setTunnelNetworkSettings(settings) { [weak self] error in
        guard let self = self else { return }

        if let error = error {
            os_log("Failed to set tunnel settings: %{public}
    @",
                   log: self.logger, type: .error,
    error.localizedDescription)
            completionHandler(error)
            return
        }

        // Start the Rust tunnel core
        do {
            try self.startRustTunnel(config: config)
            self.isTunnelActive = true
            self.saveStateToAppGroup(state: .connected,
    config: config)
            completionHandler(nil)
        } catch {
            os_log("Failed to start Rust tunnel: %{public}@",
                   log: self.logger, type: .error,
    error.localizedDescription)
            completionHandler(error)
        }
    }
}

override func stopTunnel(
    with reason: NEProviderStopReason,
    completionHandler: @escaping () -> Void
```

```
) {
    os_log("Stopping tunnel, reason: %d", log: logger,
    type: .info, reason.rawValue)

    isTunnelActive = false
    stopRustTunnel()
    saveStateToAppGroup(state: .disconnected, config: nil)

    completionHandler()
}

// MARK: - App Extension IPC

override func handleAppMessage(
    _ messageData: Data,
    completionHandler: ((Data?) -> Void)?
) {
    // Handle IPC messages from the main app
    guard let message = try? JSONSerialization.jsonObject(
        with: messageData, options: []
    ) as? [String: Any] else {
        completionHandler?(nil)
        return
    }

    let action = message["action"] as? String ?? ""

    switch action {
    case "getStats":
        let stats = getRustTunnelStats()
        let responseData = try?
    JSONSerialization.data(withJSONObject: stats)
        completionHandler?(responseData)

    case "getStatus":
        let status: [String: Any] = [
            "state": isTunnelActive ? "connected" :
    "disconnected",
            "timestamp": Date().timeIntervalSince1970
        ]
        let responseData = try?
    JSONSerialization.data(withJSONObject: status)
        completionHandler?(responseData)

    default:
        completionHandler?(nil)
    }
}

// MARK: - Sleep / Wake Handling

override func sleep(completionHandler: @escaping () -> Void) {
```

```
        // Save any necessary state before device sleeps
        os_log("Device going to sleep", log: logger, type: .info)
        // Rust WireGuard core handles sleep silently (no
        keepalives)
        completionHandler()
    }

    override func wake() {
        // Restore state after device wakes
        os_log("Device waking up", log: logger, type: .info)
        if isTunnelActive {
            // Trigger a keepalive to verify tunnel is still alive
            rustTriggerKeepalive()
        }
    }

    // MARK: - Private Methods

    private func buildTunnelNetworkSettings(config: TunnelConfig)
        -> NEPacketTunnelNetworkSettings {
        let settings = NEPacketTunnelNetworkSettings(
            tunnelRemoteAddress: config.serverAddress
        )

        // IPv4 configuration
        let ipv4Settings = NEIPv4Settings(
            addresses: [config.tunnelIPv4Address],
            subnetMasks: [config.tunnelIPv4SubnetMask]
        )

        if config.routeAllTraffic {
            ipv4Settings.includedRoutes = [NEIPv4Route.default()]
        } else {
            ipv4Settings.includedRoutes =
        config.includedRoutes.map { route in
                NEIPv4Route(
                    destinationAddress: route.address,
                    subnetMask: route.subnetMask
                )
            }
        }

        // Exclude routes (split tunneling)
        ipv4Settings.excludedRoutes = config.excludedRoutes.map {
        route in
            NEIPv4Route(
                destinationAddress: route.address,
                subnetMask: route.subnetMask
            )
        }

        settings.ipv4Settings = ipv4Settings
```

```
        // IPv6 configuration (if enabled)
        if config.enableIPv6 {
            let ipv6Settings = NEIPv6Settings(
                addresses: [config.tunnelIPv6Address],
                networkPrefixLengths: [NSNumber(value:
        config.tunnelIPv6PrefixLength)]
            )
            ipv6Settings.includedRoutes = [NEIPv6Route.default()]
            settings.ipv6Settings = ipv6Settings
        }

        // DNS configuration (critical for leak prevention)
        let dnsSettings = NEDNSSettings(servers:
        config.dnsServers)
        dnsSettings.matchDomains = [""] // Use VPN DNS for all
        queries
        settings.dnsSettings = dnsSettings

        // MTU
        settings.mtu = NSNumber(value: config.mtu)

        return settings
    }

    private func startRustTunnel(config: TunnelConfig) throws {
        // Convert config to JSON string for Rust core
        let configJson = try JSONEncoder().encode(config)
        guard let configString = String(data: configJson,
        encoding: .utf8) else {
            throw TunnelError.invalidConfiguration
        }

        // Call into Rust core via FFI
        // The Rust core receives the TUN fd internally through
        the NEPacketTunnelProvider
        let result = helix_core_start_tunnel(configString)

        if result < 0 {
            throw TunnelError.rustCoreError(code: result)
        }

        rustTunnelHandle = OpaquePointer(bitPattern: result)
    }

    private func stopRustTunnel() {
        if let handle = rustTunnelHandle {
            helix_core_stop_tunnel(handle)
            rustTunnelHandle = nil
        }
    }

    private func getRustTunnelStats() -> [String: Any] {
```

```
        guard let handle = rustTunnelHandle else {
            return [:]
        }

        let statsJson = helix_core_get_stats(handle)
        defer { free(statsJson) }

        guard let jsonData = String(cString:
        statsJson).data(using: .utf8),
              let stats = try? JSONSerialization.jsonObject(with:
        jsonData) as? [String: Any] else {
            return [:]
        }

        return stats
    }

    private func rustTriggerKeepalive() {
        guard let handle = rustTunnelHandle else { return }
        helix_core_trigger_keepalive(handle)
    }

    private func loadConfiguration(from providerConfig: [String:
        Any]) -> TunnelConfig {
        // Load configuration from providerConfiguration
        dictionary
        // and merge with App Group shared preferences
        var config = TunnelConfig()

        if let wgConfig = providerConfig["wgQuickConfig"] as?
        String {
            config.wireGuardConfig = wgConfig
        }
        if let serverAddress = providerConfig["serverAddress"]
        as? String {
            config.serverAddress = serverAddress
        }

        // Load from App Group UserDefaults
        if let sharedDefaults = UserDefaults(suiteName:
        "group.com.helix.vpn") {
            config.dnsServers =
        sharedDefaults.stringArray(forKey: "dns_servers")
                ?? ["1.1.1.1", "8.8.8.8"]
        }

        return config
    }

    private func saveStateToAppGroup(state: TunnelState, config:
        TunnelConfig?) {
        guard let sharedDefaults = UserDefaults(suiteName:
        "group.com.helix.vpn") else { return }
```

```
        sharedDefaults.set(state.rawValue, forKey: "tunnel_state")
        sharedDefaults.set(Date().timeIntervalSince1970, forKey:
        "last_state_update")

        if let config = config {
            sharedDefaults.set(config.serverAddress, forKey:
        "last_server")
        }

        sharedDefaults.synchronize()
    }

    enum TunnelState: String {
        case connected, disconnected, connecting, error
    }

    enum TunnelError: Error {
        case invalidConfiguration
        case rustCoreError(code: Int32)
        case memoryLimitExceeded
    }
}
```

3.2.2 Memory Constraint Management

The iOS Network Extension has a strict ~15MB memory limit. Every byte counts:

```
// Memory management utilities for PacketTunnelProvider
import os.log

extension PacketTunnelProvider {

    /// Log current memory usage for debugging (remove in
        production)
    func logMemoryUsage() {
        var info = mach_task_basic_info()
        var count =
        mach_msg_type_number_t(MemoryLayout<mach_task_basic_info>.size)/
        4

        let kerr: kern_return_t = withUnsafeMutablePointer(to:
        &info) {
            $0.withMemoryRebound(to: integer_t.self, capacity: 1)
        {
                task_info(mach_task_self_,
        task_flavor_t(MACH_TASK_BASIC_INFO), $0, &count)
            }
        }

        if kerr == KERN_SUCCESS {
```

```
            let usedMB = Double(info.resident_size) / 1024.0 /
        1024.0
            os_log("Memory usage: %.2f MB", log: logger,
        type: .debug, usedMB)
        }
    }

    /// Optimization strategies for the 15MB limit:
    /// 1. Keep Swift wrapper minimal - all protocol logic in Rust
    /// 2. Use structs instead of classes where possible
    /// 3. Avoid closures that capture self strongly
    /// 4. Use autoreleasepool for tight loops
    /// 5. Disable logging in release builds
    /// 6. Use compressed data structures

    func optimizedPacketLoop() {
        autoreleasepool {
            // Process packets within autoreleasepool to prevent
            // accumulation of autoreleased objects
        }
    }
}
```

Memory Budget Allocation (15MB total): | Component | Budget | Notes | |-----------|--------|-------| | Rust WireGuard core | ~6-8 MB | GotaTun / BoringTun optimized build | | Swift runtime + NEPacketTunnelProvider | ~2-3 MB | Framework overhead | | Network buffers | ~2 MB | Packet I/O buffers | | Connection state | ~0.5 MB | Peer tracking, timers | | Logging (debug only) | ~0.5 MB | Strip in release | | Reserve | ~2-3 MB | Safety margin |

3.2.3 On-Demand Rule Configuration

```
import NetworkExtension

class OnDemandConfigurator {

    /// Configure VPN On-Demand rules for automatic connection
    static func setupOnDemandRules(
        manager: NETunnelProviderManager,
        config: OnDemandConfig
    ) {
        var rules: [NEOnDemandRule] = []

        // Rule 1: Connect on untrusted Wi-Fi networks
        if config.connectOnUntrustedWiFi {
            let wifiRule = NEOnDemandRuleConnect()
            wifiRule.interfaceTypeMatch = .wiFi
            wifiRule.ssidMatch = config.trustedWiFiNetworks
            wifiRule.action = .connect
            rules.append(wifiRule)
        }
```

```
        // Rule 2: Connect on cellular
        if config.connectOnCellular {
            let cellularRule = NEOnDemandRuleConnect()
            cellularRule.interfaceTypeMatch = .cellular
            cellularRule.action = .connect
            rules.append(cellularRule)
        }

        // Rule 3: Connect on any Ethernet (for iPad with dongle)
        if config.connectOnEthernet {
            let ethernetRule = NEOnDemandRuleConnect()
            ethernetRule.interfaceTypeMatch = .ethernet
            ethernetRule.action = .connect
            rules.append(ethernetRule)
        }

        // Rule 4: Disconnect on trusted Wi-Fi (e.g., home
        network)
        if !config.trustedWiFiNetworks.isEmpty {
            let trustedWifiRule = NEOnDemandRuleDisconnect()
            trustedWifiRule.interfaceTypeMatch = .wiFi
            trustedWifiRule.ssidMatch = config.trustedWiFiNetworks
            trustedWifiRule.action = .disconnect
            rules.append(trustedWifiRule)
        }

        // Default rule: ignore (do nothing)
        let defaultRule = NEOnDemandRuleIgnore()
        rules.append(defaultRule)

        manager.onDemandRules = rules
        manager.isOnDemandEnabled = config.enabled
    }

    /// Domain-based on-demand for split tunneling evaluation
    static func setupDomainEvaluationRules(
        manager: NETunnelProviderManager,
        domains: [String]
    ) {
        let evaluationRule = NEOnDemandRuleEvaluateConnection()
        evaluationRule.interfaceTypeMatch = .any

        let connectionRules = domains.map { domain in
            NEEvaluateConnectionRule(
                matchDomains: [domain],
                andAction: .connectIfNeeded
            )
        }

        evaluationRule.connectionRules = connectionRules
        manager.onDemandRules = [evaluationRule]
```

```
        manager.isOnDemandEnabled = true
    }
}

struct OnDemandConfig {
    var enabled: Bool = false
    var connectOnUntrustedWiFi: Bool = true
    var connectOnCellular: Bool = false
    var connectOnEthernet: Bool = true
    var trustedWiFiNetworks: [String] = []
}
```

3.3 App Groups for Container Sharing

```
// Shared container between main app and Network Extension
class AppGroupContainer {
    static let shared = AppGroupContainer()
    static let suiteName = "group.com.helix.vpn"

    private let userDefaults: UserDefaults?
    private let fileManager = FileManager.default

    private init() {
        userDefaults = UserDefaults(suiteName:
        AppGroupContainer.suiteName)
    }

    // MARK: - UserDefaults helpers

    func set(_ value: Any?, forKey key: String) {
        userDefaults?.set(value, forKey: key)
    }

    func string(forKey key: String) -> String? {
        return userDefaults?.string(forKey: key)
    }

    func bool(forKey key: String) -> Bool {
        return userDefaults?.bool(forKey: key) ?? false
    }

    // MARK: - File sharing

    var sharedContainerURL: URL? {
        return
        fileManager.containerURL(forSecurityApplicationGroupIdentifier:
            AppGroupContainer.suiteName)
    }

    func saveConfigurationFile(_ data: Data, filename: String)
        throws {
        guard let containerURL = sharedContainerURL else {
```

```
            throw AppGroupError.containerNotAvailable
        }

        let fileURL =
        containerURL.appendingPathComponent(filename)
        try data.write(to: fileURL, options: .atomic)
    }

    func loadConfigurationFile(filename: String) -> Data? {
        guard let containerURL = sharedContainerURL else { return
        nil }
        let fileURL =
        containerURL.appendingPathComponent(filename)
        return try? Data(contentsOf: fileURL)
    }

    enum AppGroupError: Error {
        case containerNotAvailable
        case writeFailed
        case readFailed
    }
}
```

3.4 Network Extension Entitlement

```
<!-- Runner/Runner.entitlements (Main App) -->
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN"
  "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
    <!-- App Groups for main app and extension sharing -->
    <key>com.apple.security.application-groups</key>
    <array>
        <string>group.com.helix.vpn</string>
    </array>


        <!-- Network Extension entitlement (self-serve since Nov
        2016) -->
    <key>com.apple.developer.networking.networkextension</key>
    <array>
        <string>packet-tunnel-provider</string>
    </array>

    <!-- Keychain sharing -->
    <key>keychain-access-groups</key>
    <array>
        <string>$(AppIdentifierPrefix)com.helix.vpn</string>
    </array>
```

```
</dict>
</plist>

<!-- PacketTunnelProvider/PacketTunnelProvider.entitlements
        (Extension) -->
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN"
  "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
    <key>com.apple.security.application-groups</key>
    <array>
        <string>group.com.helix.vpn</string>
    </array>
</dict>
</plist>
```

3.5 Keychain Integration for Credentials

```
import Security
import LocalAuthentication

class KeychainCredentialStore {

    static let shared = KeychainCredentialStore()
    private let service = "com.helix.vpn.credentials"

    /// Save VPN credentials with optional biometric protection
    func saveCredentials(
        account: String,
        password: String,
        biometricRequired: Bool = false
    ) -> OSStatus {
        // Delete existing item first
        deleteCredentials(account: account)

        var query: [String: Any] = [
            kSecClass as String: kSecClassGenericPassword,
            kSecAttrAccount as String: account,
            kSecAttrService as String: service,
            kSecValueData as String: password.data(using: .utf8)!,
            kSecAttrAccessible as String:
        kSecAttrAccessibleWhenUnlockedThisDeviceOnly
        ]

        // Add biometric protection if requested
        if biometricRequired {
            var error: Unmanaged<CFError>?
            guard let accessControl =
        SecAccessControlCreateWithFlags(
                kCFAllocatorDefault,
                kSecAttrAccessibleWhenUnlockedThisDeviceOnly,
```

```
            .biometryCurrentSet,
            &error
        ) else {
            return errSecParam
        }
        query[kSecAttrAccessControl as String] = accessControl
        // Remove accessible attribute when using access
    control
        query.removeValue(forKey: kSecAttrAccessible as
    String)
    }

    return SecItemAdd(query as CFDictionary, nil)
}

/// Retrieve credentials (triggers biometric prompt if
    protected)
func retrieveCredentials(account: String) -> String? {
    let query: [String: Any] = [
        kSecClass as String: kSecClassGenericPassword,
        kSecAttrAccount as String: account,
        kSecAttrService as String: service,
        kSecReturnData as String: true,
        kSecMatchLimit as String: kSecMatchLimitOne,
        kSecUseOperationPrompt as String: "Authenticate to
    access VPN credentials"
    ]

    var result: CFTypeRef?
    let status = SecItemCopyMatching(query as CFDictionary,
    &result)

    guard status == errSecSuccess,
          let data = result as? Data,
          let password = String(data: data, encoding: .utf8)
    else {
        return nil
    }

    return password
}

/// Delete stored credentials
func deleteCredentials(account: String) -> OSStatus {
    let query: [String: Any] = [
        kSecClass as String: kSecClassGenericPassword,
        kSecAttrAccount as String: account,
        kSecAttrService as String: service
    ]
    return SecItemDelete(query as CFDictionary)
}

/// Save WireGuard private key securely
```

```
    func saveWireGuardPrivateKey(_ privateKey: String) ->
        OSStatus {
        return saveCredentials(account: "wg_private_key",
        password: privateKey)
    }

    /// Retrieve WireGuard private key
    func getWireGuardPrivateKey() -> String? {
        return retrieveCredentials(account: "wg_private_key")
    }
}
```

3.6 Control Center Integration

iOS VPN apps automatically appear in Settings > VPN and can be toggled via Control Center. The app provides the configuration; the system handles the UI:

```
import NetworkExtension

class VpnConfigurationManager {

    static let shared = VpnConfigurationManager()

    /// Load or create the VPN configuration
    func loadOrCreateConfiguration(
        completion: @escaping (NETunnelProviderManager?, Error?)
        -> Void
    ) {
        NETunnelProviderManager.loadAllFromPreferences {
        managers, error in
            if let error = error {
                completion(nil, error)
                return
            }

            // Return existing configuration if found
            if let existingManager = managers?.first(where: {
        manager in
                (manager.protocolConfiguration as?
        NETunnelProviderProtocol)?
                    .providerBundleIdentifier ==
        "com.helix.vpn.packet-tunnel"
            }) {
                completion(existingManager, nil)
                return
            }

            // Create new configuration
            let newManager = self.createNewConfiguration()
            newManager.saveToPreferences { error in
                if let error = error {
```

```
                completion(nil, error)
                return
            }
            // Must reload after saving
            NETunnelProviderManager.loadAllFromPreferences {
    _, _ in
                completion(newManager, nil)
            }
        }
    }
}

private func createNewConfiguration() ->
    NETunnelProviderManager {
    let manager = NETunnelProviderManager()

    let protocolConfig = NETunnelProviderProtocol()
    protocolConfig.providerBundleIdentifier =
    "com.helix.vpn.packet-tunnel"
    protocolConfig.serverAddress = "helix-vpn.example.com"
    protocolConfig.providerConfiguration = [
        "wgQuickConfig": "", // Populated at connection time
        "protocol": "wireguard"
    ]

    manager.protocolConfiguration = protocolConfig
    manager.localizedDescription = "Helix VPN"
    manager.isEnabled = true

    return manager
}

/// Start the VPN tunnel
func connect(
    manager: NETunnelProviderManager,
    config: TunnelConfig,
    completion: @escaping (Error?) -> Void
) {
    // Update protocol configuration with current server
    if let proto = manager.protocolConfiguration as?
    NETunnelProviderProtocol {
        var providerConfig = proto.providerConfiguration ??
    [:]
        providerConfig["serverAddress"] =
    config.serverAddress as NSString
        providerConfig["wgQuickConfig"] =
    config.wireGuardConfig as NSString
        proto.providerConfiguration = providerConfig
        proto.serverAddress = config.serverAddress
    }

    manager.saveToPreferences { error in
        if let error = error {
```

```
                completion(error)
                return
            }

            do {
                try manager.connection.startVPNTunnel()
                completion(nil)
            } catch {
                completion(error)
            }
        }
    }

    /// Stop the VPN tunnel
    func disconnect(manager: NETunnelProviderManager) {
        manager.connection.stopVPNTunnel()
    }

    /// Monitor connection status
    var connectionStatus: NEVPNStatus {
        //
        Returns: .invalid, .disconnected, .connecting, .connected, .reasserting,
        return NETunnelProviderManager().connection.status
    }
}
```

3.7 iOS-Specific UI Adaptations

```
// lib/ui/platform/ios_ui_adapters.dart
import 'package:flutter/cupertino.dart';
import 'package:flutter/material.dart';
import 'dart:io' show Platform;

/// iOS-specific UI adaptations to match platform conventions
class IOSUiAdapter {

  /// Use Cupertino-style navigation bar on iOS
  static PreferredSizeWidget buildAppBar({
    required BuildContext context,
    required String title,
    List<Widget>? actions,
    Widget? leading,
    bool centerTitle = true,
  }) {
    if (Platform.isIOS) {
      return CupertinoNavigationBar(
        middle: Text(title),
        trailing: actions != null && actions.isNotEmpty
            ? Row(mainAxisSize: MainAxisSize.min, children:
        actions)
            : null,
```

```
      leading: leading,
      backgroundColor:
      CupertinoColors.systemBackground.withOpacity(0.9),
    ) as PreferredSizeWidget;
  }

  return AppBar(
    title: Text(title),
    centerTitle: centerTitle,
    actions: actions,
    leading: leading,
  );
}

/// Platform-appropriate button style
static Widget buildPrimaryButton({
  required String label,
  required VoidCallback onPressed,
  bool isDestructive = false,
}) {
  if (Platform.isIOS) {
    return CupertinoButton.filled(
      onPressed: onPressed,
      child: Text(label),
    );
  }

  return ElevatedButton(
    onPressed: onPressed,
    style: isDestructive
        ? ElevatedButton.styleFrom(backgroundColor: Colors.red)
        : null,
    child: Text(label),
  );
}

/// Platform-appropriate settings list
static Widget buildSettingsList({
  required List<SettingsItem> items,
}) {
  if (Platform.isIOS) {
    return CupertinoFormSection.insetGrouped(
      children: items.map((item) => CupertinoFormRow(
        prefix: Row(
          children: [
            if (item.icon != null)
              Padding(
                padding: const EdgeInsets.only(right: 12),
                child: Icon(item.icon, color:
      CupertinoColors.systemBlue),
              ),
            Text(item.title),
```

```
          ],
        ),
        child: item.trailing ?? const SizedBox.shrink(),
        helper: item.subtitle != null
            ? Text(item.subtitle!, style: const
      TextStyle(fontSize: 12))
            : null,
      )).toList(),
    );
  }

  return ListView.builder(
    itemCount: items.length,
    itemBuilder: (context, index) {
      final item = items[index];
      return ListTile(
        leading: item.icon != null ? Icon(item.icon) : null,
        title: Text(item.title),
        subtitle: item.subtitle != null ?
      Text(item.subtitle!) : null,
        trailing: item.trailing ?? const
      Icon(Icons.chevron_right),
        onTap: item.onTap,
      );
    },
  );
}

/// Platform-appropriate connection toggle (large central
      button)
static Widget buildConnectionToggle({
  required bool isConnected,
  required VoidCallback onToggle,
  required bool isConnecting,
}) {
  if (Platform.isIOS) {
    return CupertinoButton(
      onPressed: isConnecting ? null : onToggle,
      child: Container(
        width: 120,
        height: 120,
        decoration: BoxDecoration(
          shape: BoxShape.circle,
          color: isConnected
              ? CupertinoColors.activeGreen
              : CupertinoColors.systemGrey,
          boxShadow: [
            BoxShadow(
              color: (isConnected
                  ? CupertinoColors.activeGreen
                  :
      CupertinoColors.systemGrey).withOpacity(0.3),
              blurRadius: 20,
```

```
                spreadRadius: 5,
              ),
            ],
          ),
          child: Center(
            child: isConnecting
                ? const CupertinoActivityIndicator(color:
        CupertinoColors.white)
                : Icon(
                    isConnected ? CupertinoIcons.check_mark :
        CupertinoIcons.power,
                    color: CupertinoColors.white,
                    size: 48,
                  ),
          ),
        ),
      );
    }

    // Material design fallback
    return FloatingActionButton.large(
      onPressed: isConnecting ? null : onToggle,
      backgroundColor: isConnected ? Colors.green : Colors.grey,
      child: isConnecting
          ? const CircularProgressIndicator(color: Colors.white)
          : Icon(isConnected ? Icons.check :
        Icons.power_settings_new, size: 48),
    );
  }
}

class SettingsItem {
  final String title;
  final String? subtitle;
  final IconData? icon;
  final Widget? trailing;
  final VoidCallback? onTap;

  SettingsItem({
    required this.title,
    this.subtitle,
    this.icon,
    this.trailing,
    this.onTap,
  });
}
```

3.8 App Store Submission Requirements

3.8.1 VPN App Review Guidelines

Apple scrutinizes VPN apps heavily. Requirements include:

- **Privacy Policy**: Must accurately describe all data collection, use, and retention practices. Must be accessible from both the App Store listing and within the app.
- **Network Extension Justification**: The app must demonstrate a legitimate use case for the Network Extension entitlement.
- **User Authorization**: User must explicitly approve the VPN configuration on first use (system-enforced).
- **Battery Impact**: Apps that drain battery excessively may be rejected. WireGuard typically passes due to its efficient design.
- **No Misleading Claims**: Avoid terms like "military-grade encryption" without substantiation.
- **App Tracking Transparency (ATT)**: If collecting any tracking data, ATT framework must be implemented.

3.8.2 Required Entitlements and Provisioning

Requirement	Details
Developer Account	Apple Developer Program ($99/year)
Entitlement	`com.apple.developer.networking.networkextension` with `packet-tunnel-provider`
App Groups	`group.com.helix.vpn` for app/extension data sharing
Provisioning Profile	Must include Network Extension entitlement
Signing	Distribution certificate for App Store

3.8.3 Info.plist Requirements

```
<!-- iOS Runner/Info.plist additions -->
<key>NSLocalNetworkUsageDescription</key>
<string>Helix VPN needs to manage your network connection to
        establish a secure VPN tunnel.</string>
<key>UIBackgroundModes</key>
<array>
    <string>network-authentication</string>
</array>
<key>NSFaceIDUsageDescription</key>
<string>Helix VPN uses Face ID to protect your VPN credentials.</
        string>
```

3.9 TestFlight Distribution

```
# Build iOS release archive
flutter build ipa --release --export-options-plist=ios/
        exportOptions.plist

# Upload to App Store Connect / TestFlight
xcrun altool --upload-app \
  --type ios \
  --file build/ios/ipa/HelixVPN.ipa \
  --apiKey "YOUR_API_KEY" \
  --apiIssuer "YOUR_ISSUER_ID"

# exportOptions.plist
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN"
  "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
    <key>method</key>
    <string>app-store</string>
    <key>teamID</key>
    <string>YOUR_TEAM_ID</string>
    <key>uploadSymbols</key>
    <true/>
    <key>uploadBitcode</key>
    <false/>
</dict>
</plist>
```

---

4. HarmonyOS Client

4.1 Architecture

The HarmonyOS client follows a similar pattern to Android but uses HarmonyOS-specific APIs:

```
┌─────────────────────────────────────────────┐
│         Flutter UI Layer (Dart)             │
│  - Flutter 3.22.0-ohos embedding layer      │
├─────────────────────────────────────────────┤
│      flutter_rust_bridge Layer              │
│  - Dart FFI bindings to Rust core           │
├─────────────────────────────────────────────┤
│     HarmonyOS Native Layer (ArkTS)          │
│  - VpnExtensionAbility                      │
│  - N-API bridge to Rust                     │
│  - Form card (service widget)               │
├─────────────────────────────────────────────┤
```

```
│         Shared Rust Core                          │
│  - helix-core via N-API                           │
│  - WireGuard/OpenVPN protocol                     │
└──────────────────────────────────────────────────┘
```

4.2 HarmonyOS VPN API (VpnExtensionAbility)

```typescript
// entry/src/main/ets/vpnability/HelixVpnAbility.ets
import { VpnExtensionAbility } from '@kit.NetworkKit';
import { Want } from '@kit.AbilityKit';
import vpn from '@ohos.net.vpn';
import hilog from '@ohos.hilog';

// N-API imports for Rust core
import { HelixRustBridge } from '../rust/HelixRustBridge';

const TAG = 'HelixVpnAbility';

export default class HelixVpnAbility extends VpnExtensionAbility {
  private vpnConnection: vpn.VpnConnection | null = null;
  private rustBridge: HelixRustBridge | null = null;
  private tunFd: number = -1;

  onCreate(want: Want): void {
    hilog.info(0x0000, TAG, 'HelixVpnAbility onCreate');

    // Initialize Rust bridge via N-API
    this.rustBridge = new HelixRustBridge();

    // Extract configuration from want parameters
    const config = want.parameters as VpnConfig;
    if (config) {
      this.setupVpn(config);
    }
  }

  onDestroy(): void {
    hilog.info(0x0000, TAG, 'HelixVpnAbility onDestroy');
    this.teardownVpn();
  }

  private async setupVpn(config: VpnConfig): Promise<void> {
    try {
      // Create VPN connection through system API
      this.vpnConnection = vpn.createVpnConnection(this.context);

      // Build VPN configuration
      const vpnConfig: vpn.VpnConfig = {
        addresses: [{
          address: { address: config.tunnelAddress, family:
        1 }, // AF_INET
```

```
        prefixLength: config.tunnelPrefixLength
      }],
      mtu: config.mtu || 1400,
      dnsAddresses: config.dnsServers,
      routes: config.routes.map(route => ({
        address: { address: route.address, family:
      route.family },
        prefixLength: route.prefixLength
      })),
      isBlocking: config.killSwitchEnabled || false, // Kill
      switch
      isIPv4Accepted: true,
      isIPv6Accepted: config.enableIPv6 || false,
      trustedApplications: config.allowedApplications || [],
      blockedApplications: config.blockedApplications || [],
    };

    // Establish VPN tunnel (returns TUN file descriptor)
    this.tunFd = await this.vpnConnection.setUp(vpnConfig);
    hilog.info(0x0000, TAG, `VPN setUp success, tunFd: $
      {this.tunFd}`);

    // Pass TUN fd to Rust core for packet processing
    if (this.rustBridge) {
      this.rustBridge.startTunnel(this.tunFd,
      JSON.stringify(config));
    }

    // Save connection state
    this.saveConnectionState('connected', config);

  } catch (error) {
    hilog.error(0x0000, TAG, `VPN setup failed: $
      {JSON.stringify(error)}`);
    this.saveConnectionState('error', config);
  }
}

private async teardownVpn(): Promise<void> {
  // Stop Rust core
  if (this.rustBridge) {
    this.rustBridge.stopTunnel();
  }

  // Destroy VPN connection
  if (this.vpnConnection) {
    try {
      await this.vpnConnection.destroy();
      hilog.info(0x0000, TAG, 'VPN connection destroyed');
    } catch (error) {
      hilog.error(0x0000, TAG, `VPN destroy error: $
      {JSON.stringify(error)}`);
    }
```

```
      this.vpnConnection = null;
    }

    this.tunFd = -1;
    this.saveConnectionState('disconnected', null);
  }

  private saveConnectionState(state: string, config: VpnConfig |
        null): void {
    // Use AppStorage or preferences for state persistence
    // so the main app can read the current VPN status
  }
}

// VpnConfig interface
interface VpnConfig {
  tunnelAddress: string;
  tunnelPrefixLength: number;
  mtu?: number;
  dnsServers: string[];
  routes: RouteConfig[];
  killSwitchEnabled?: boolean;
  enableIPv6?: boolean;
  allowedApplications?: string[];
  blockedApplications?: string[];
}

interface RouteConfig {
  address: string;
  family: number;
  prefixLength: number;
}
```

4.3 N-API Bridge for Rust

```
// entry/src/main/ets/rust/HelixRustBridge.ets
// N-API bridge module declaration
import napi from 'libnapi.so'; // Native library

const TAG = 'HelixRustBridge';

/**
 * Bridge to helix-core Rust library via N-API.
 * This is the HarmonyOS equivalent of JNI (Android) or C FFI
       (iOS).
 */
export class HelixRustBridge {
  private nativeHandle: napi.NativeHandle;

  constructor() {
    // Initialize the native Rust module
```

```
    this.nativeHandle = napi.initHelixCore();
  }

  /**
   * Start the VPN tunnel with the given TUN fd and configuration.
   * @param tunFd - File descriptor of the TUN device
   * @param configJson - JSON string containing tunnel
       configuration
   */
  startTunnel(tunFd: number, configJson: string): number {
    return napi.helixStartTunnel(this.nativeHandle, tunFd,
        configJson);
  }

  /**
   * Stop the active VPN tunnel.
   */
  stopTunnel(): number {
    return napi.helixStopTunnel(this.nativeHandle);
  }

  /**
   * Get current connection statistics.
   */
  getStats(): ConnectionStats {
    const statsJson = napi.helixGetStats(this.nativeHandle);
    return JSON.parse(statsJson) as ConnectionStats;
  }

  /**
   * Get current tunnel state.
   */
  getState(): TunnelState {
    return napi.helixGetState(this.nativeHandle) as TunnelState;
  }

  /**
   * Release native resources.
   */
  destroy(): void {
    napi.helixDestroy(this.nativeHandle);
  }
}

interface ConnectionStats {
  uploadSpeed: number;
  downloadSpeed: number;
  totalUpload: number;
  totalDownload: number;
  duration: number;
}
```

```
type TunnelState = 'disconnected' | 'connecting' | 'connected' |
        'disconnecting' | 'error';
```

Native C++ N-API binding layer:

```
// entry/src/main/cpp/napi_helix_bridge.cpp
#include <napi/native_api.h>
#include <hilog/log.h>
#include <string>
#include "helix_core.h" // Rust-generated C header

#define LOG_TAG "HelixNAPI"

// Forward declarations of Rust FFI functions
extern "C" {
    void* helix_core_init();
    int32_t helix_start_tunnel(void* handle, int32_t tun_fd,
        const char* config_json);
    int32_t helix_stop_tunnel(void* handle);
    const char* helix_get_stats(void* handle);
    int32_t helix_get_state(void* handle);
    void helix_destroy(void* handle);
    void helix_free_string(const char* str);
}

static napi_value NAPI_InitHelixCore(napi_env env,
        napi_callback_info info) {
    void* handle = helix_core_init();
    napi_value result;
    napi_create_bigint_uint64(env,
        reinterpret_cast<uint64_t>(handle), &result);
    return result;
}

static napi_value NAPI_HelixStartTunnel(napi_env env,
        napi_callback_info info) {
    size_t argc = 3;
    napi_value args[3] = {nullptr};
    napi_get_cb_info(env, info, &argc, args, nullptr, nullptr);

    // Extract handle
    uint64_t handle_val;
    napi_get_value_bigint_uint64(env, args[0], &handle_val,
        nullptr);
    void* handle = reinterpret_cast<void*>(handle_val);

    // Extract TUN fd
    int32_t tun_fd;
    napi_get_value_int32(env, args[1], &tun_fd);

    // Extract config JSON
    size_t config_len;
```

```
    napi_get_value_string_utf8(env, args[2], nullptr, 0,
        &config_len);
    std::string config_json(config_len, '\0');
    napi_get_value_string_utf8(env, args[2], &config_json[0],
        config_len + 1, &config_len);

    int32_t result = helix_start_tunnel(handle, tun_fd,
        config_json.c_str());

    napi_value napi_result;
    napi_create_int32(env, result, &napi_result);
    return napi_result;
}

static napi_value NAPI_HelixGetStats(napi_env env,
        napi_callback_info info) {
    size_t argc = 1;
    napi_value args[1] = {nullptr};
    napi_get_cb_info(env, info, &argc, args, nullptr, nullptr);

    uint64_t handle_val;
    napi_get_value_bigint_uint64(env, args[0], &handle_val,
        nullptr);
    void* handle = reinterpret_cast<void*>(handle_val);

    const char* stats = helix_get_stats(handle);
    napi_value result;
    napi_create_string_utf8(env, stats, NAPI_AUTO_LENGTH,
        &result);
    helix_free_string(stats);

    return result;
}

// Module registration
static napi_value Init(napi_env env, napi_value exports) {
    napi_property_descriptor desc[] = {
        {"initHelixCore", nullptr, NAPI_InitHelixCore, nullptr,
        nullptr, nullptr, napi_default, nullptr},
        {"helixStartTunnel", nullptr, NAPI_HelixStartTunnel,
        nullptr, nullptr, nullptr, napi_default, nullptr},
        {"helixGetStats", nullptr, NAPI_HelixGetStats, nullptr,
        nullptr, nullptr, napi_default, nullptr},
        // ... other methods
    };
    napi_define_properties(env, exports, sizeof(desc) /
        sizeof(desc[0]), desc);
    return exports;
}

static napi_module helixModule = {
    .nm_version = 1,
    .nm_flags = 0,
```

```
    .nm_filename = nullptr,
    .nm_register_func = Init,
    .nm_modname = "libnapi",
    .nm_priv = nullptr,
    .reserved = {0}
};

extern "C" __attribute__((constructor)) void RegisterModule() {
    napi_module_register(&helixModule);
}
```

4.4 DevEco Studio Build Setup

```
// build-profile.json5
{
  "app": {
    "signingConfigs": [],
    "compileSdkVersion": 12,
    "compatibleSdkVersion": 12,
    "products": [
      {
        "name": "default",
        "signingConfig": "default",
        "compatibleSdkVersion": "12",
        "runtimeOS": "HarmonyOS"
      }
    ],
    "buildOption": {
      "strictMode": {
        "caseSensitiveCheck": true,
        "useNormalizedOHMUrl": true
      }
    }
  },
  "modules": [
    {
      "name": "entry",
      "srcPath": "./entry",
      "targets": [
        {
          "name": "default",
          "applyToProducts": ["default"]
        }
      ]
    }
  ]
}

// entry/build-profile.json5
{
  "apiType": "stageMode",
```

```
  "buildOption": {
    "externalOptions": {
      "ignore ossAudit": true
    },
    "arkOptions": {
      "buildProfileFields": {
        "HARMONY_OS_BUILD": "true"
      }
    },
    "nativeOptions": {
      "libraries": [
        {
          "name": "libnapi.so",
          "path": "./src/main/cpp/",
          "napi": true
        }
      ]
    }
  },
  "buildOptionSet": [
    {
      "name": "release",
      "arkOptions": {
        "obfuscation": {
          "ruleOptions": {
            "enable": true,
            "files": ["./obfuscation-rules.txt"]
          }
        }
      }
    }
  ],
  "targets": [
    {
      "name": "default"
    }
  ]
}

// entry/module.json5
{
  "module": {
    "name": "entry",
    "type": "entry",
    "description": "$string:module_desc",
    "mainElement": "EntryAbility",
    "deviceTypes": ["phone", "tablet"],
    "deliveryWithInstall": true,
    "installationFree": false,
    "pages": "$profile:main_pages",
    "abilities": [
      {
```

```
        "name": "EntryAbility",
        "srcEntry": "./ets/entryability/EntryAbility.ets",
        "description": "$string:EntryAbility_desc",
        "icon": "$media:icon",
        "label": "$string:EntryAbility_label",
        "startWindowIcon": "$media:startIcon",
        "startWindowBackground": "$color:start_window_background",
        "exported": true,
        "skills": [
          {
            "entities": ["entity.system.home"],
            "actions": ["action.system.home"]
          }
        ]
      }
    ],
    "extensionAbilities": [
      {
        "name": "HelixVpnAbility",
        "srcEntry": "./ets/vpnability/HelixVpnAbility.ets",
        "type": "vpn",
        "exported": false
      }
    ],
    "requestPermissions": [
      {
        "name": "ohos.permission.MANAGE_VPN",
        "reason": "$string:permission_vpn_reason",
        "usedScene": {
          "abilities": ["HelixVpnAbility"],
          "when": "inuse"
        }
      },
      {
        "name": "ohos.permission.INTERNET"
      },
      {
        "name": "ohos.permission.GET_NETWORK_INFO"
      }
    ]
  }
}
```

4.5 ohos-specific Flutter Plugins

```yaml
# pubspec.yaml - HarmonyOS-specific dependencies
name: helix_vpn
version: 1.0.0
environment:
  sdk: '>=3.0.0 <4.0.0'
  flutter: ">=3.22.0"
```

```yaml
dependencies:
  flutter:
    sdk: flutter

  # Core dependencies (all platforms)
  flutter_rust_bridge: ^2.0.0
  riverpod: ^2.5.0
  dio: ^5.4.0
  shared_preferences: ^2.2.0

  # Conditional HarmonyOS dependencies
  connectivity_plus: ^5.0.0

dependency_overrides:
  # Use HarmonyOS-compatible versions
  path_provider:
    git:
      url: https://gitee.com/openharmony-sig/flutter_packages.git
      path: packages/path_provider/path_provider

dev_dependencies:
  flutter_test:
    sdk: flutter
  build_runner: ^2.4.0
  freezed: ^2.4.0

flutter:
  uses-material-design: true
  assets:
    - assets/images/
    - assets/flags/
```

4.6 AppGallery Distribution

```
# Build HarmonyOS HAP package
flutter build hap --release

# Output: build/ohos/outputs/default/entry-default-signed.hap

# AppGallery Connect upload
# 1. Register app at https://developer.huawei.com/consumer/en/
        appgallery/
# 2. Sign the HAP with Huawei certificates
# 3. Upload through AppGallery Connect web interface or CLI

# Signing configuration requires:
# - Huawei Developer account
# - App signing certificate (.cer)
```

```
# - App signing private key (.p12)
# - Profile file (.p7b)
```

4.7 Form Card (Service Widget) for Quick Connect

```
// entry/src/main/ets/widget/pages/HelixWidgetCard.ets
import { AppRouter } from '@kit.ArkUI';
import formBindingData from '@ohos.app.form.formBindingData';
import FormExtensionAbility from
        '@ohos.app.form.FormExtensionAbility';
import formProvider from '@ohos.app.form.formProvider';

// Widget form data
interface WidgetFormData {
  vpnStatus: string;
  serverLocation: string;
  connectButtonText: string;
  statusColor: string;
}

export default class HelixWidgetAbility extends
        FormExtensionAbility {

  onCreate(want: Want): formBindingData.FormBindingData {
    // Initial widget data
    const formData: WidgetFormData = {
      vpnStatus: 'Disconnected',
      serverLocation: 'Tap to connect',
      connectButtonText: 'Connect',
      statusColor: '#808080'
    };

    return formBindingData.createFormBindingData(formData);
  }

  onUpdate(formId: string): void {
    // Read current VPN state from preferences
    const vpnState = this.readVpnState();

    const formData: WidgetFormData = {
      vpnStatus: vpnState.connected ? 'Connected' :
        'Disconnected',
      serverLocation: vpnState.serverLocation || 'Tap to connect',
      connectButtonText: vpnState.connected ? 'Disconnect' :
        'Connect',
      statusColor: vpnState.connected ? '#00C853' : '#808080'
    };

    formProvider.updateForm(formId,
        formBindingData.createFormBindingData(formData));
  }
```

```
  onFormEvent(formId: string, message: string): void {
    const event = JSON.parse(message);

    if (event.action === 'toggle') {
      // Launch VPN ability to toggle connection
      const want: Want = {
        bundleName: 'com.helix.vpn',
        abilityName: 'HelixVpnAbility',
        parameters: {
          action: event.connected ? 'disconnect' : 'connect'
        }
      };
      this.context.startAbility(want);
    }
  }

  private readVpnState(): { connected: boolean; serverLocation?:
        string } {
    // Read from AppStorage or preferences
    return { connected: false };
  }
}

// Widget configuration in module.json5
{
  "extensionAbilities": [
    {
      "name": "HelixWidgetAbility",
      "srcEntry": "./ets/widget/HelixWidgetAbility.ets",
      "type": "form",
      "metadata": [
        {
          "name": "ohos.extension.form",
          "resource": "$profile:form_config"
        }
      ]
    }
  ]
}
```

4.8 HarmonyOS NEXT Specific Considerations

Aspect	HarmonyOS NEXT	Android
App enumeration	Cannot list all installed apps (privacy)	Full access via PackageManager
VPN permission	`MANAGE_VPN` requires ACL (API 12+)	`BIND_VPN_SERVICE` via system dialog
Emulator		Full emulator support (with some limitations)

Aspect	HarmonyOS NEXT	Android
	ARM simulator; VPN auth component may be missing	
Native bridge	N-API (Node-API based on Node.js 12.x)	JNI
Build system	DevEco Studio + hvigor	Android Studio + Gradle
Package format	HAP (HarmonyOS Ability Package)	APK / AAB
Distribution	Huawei AppGallery	Google Play, F-Droid, direct
UI framework	ArkUI (ArkTS) + Flutter embedding	Native + Flutter embedding

Key differences from Android VpnService:

1. **No global app enumeration**: HarmonyOS NEXT restricts listing all installed apps, which affects per-app split tunneling UIs. The app must use preset app lists or manual package name entry.
2. **VpnExtensionAbility vs VpnService**: HarmonyOS uses `VpnExtensionAbility` with `onCreate`/`onDestroy` callbacks instead of extending `VpnService`.
3. **N-API vs JNI**: The native bridge uses Node-API (N-API) instead of JNI, though both follow similar patterns for wrapping native objects.
4. **Permission model**: `MANAGE_VPN` was restricted to system apps until API 11; API 12+ enables third-party apps via ACL.

---

5. Mobile UI/UX Design

5.1 Flutter Design System

```dart
// lib/theme/helix_theme.dart
import 'package:flutter/material.dart';
import 'dart:io' show Platform;

class HelixTheme {
  // Brand colors
  static const Color primaryColor = Color(0xFF6366F1);
  static const Color primaryDark = Color(0xFF4F46E5);
  static const Color accentColor = Color(0xFF10B981);
  static const Color errorColor = Color(0xFFEF4444);
  static const Color warningColor = Color(0xFFF59E0B);
```

```dart
  // Connection state colors
  static const Color connected = Color(0xFF10B981);
  static const Color connecting = Color(0xFFF59E0B);
  static const Color disconnected = Color(0xFF6B7280);
  static const Color error = Color(0xFFEF4444);

  // Background colors
  static const Color darkBackground = Color(0xFF0F172A);
  static const Color darkSurface = Color(0xFF1E293B);
  static const Color lightBackground = Color(0xFFF8FAFC);
  static const Color lightSurface = Colors.white;

  // Text colors
  static const Color textPrimary = Color(0xFF1E293B);
  static const Color textSecondary = Color(0xFF64748B);
  static const Color textOnDark = Colors.white;

  /// Get theme based on platform and brightness preference
  static ThemeData getTheme({required bool isDarkMode}) {
    final baseTheme = isDarkMode ? _darkTheme : _lightTheme;

    if (Platform.isIOS) {
      return baseTheme.copyWith(
        platform: TargetPlatform.iOS,
        pageTransitionsTheme: const PageTransitionsTheme(
          builders: {
            TargetPlatform.iOS: CupertinoPageTransitionsBuilder(),
          },
        ),
      );
    }

    return baseTheme;
  }

  static final ThemeData _lightTheme = ThemeData(
    useMaterial3: true,
    brightness: Brightness.light,
    colorScheme: ColorScheme.fromSeed(
      seedColor: primaryColor,
      brightness: Brightness.light,
    ),
    scaffoldBackgroundColor: lightBackground,
    cardTheme: CardTheme(
      elevation: 0,
      shape: RoundedRectangleBorder(borderRadius:
        BorderRadius.circular(16)),
      color: lightSurface,
    ),
    appBarTheme: const AppBarTheme(
      elevation: 0,
      centerTitle: true,
```

```
      backgroundColor: lightSurface,
      foregroundColor: textPrimary,
    ),
    bottomNavigationBarTheme: const BottomNavigationBarThemeData(
      backgroundColor: lightSurface,
      selectedItemColor: primaryColor,
      unselectedItemColor: textSecondary,
      type: BottomNavigationBarType.fixed,
    ),
  );

  static final ThemeData _darkTheme = ThemeData(
    useMaterial3: true,
    brightness: Brightness.dark,
    colorScheme: ColorScheme.fromSeed(
      seedColor: primaryColor,
      brightness: Brightness.dark,
    ),
    scaffoldBackgroundColor: darkBackground,
    cardTheme: CardTheme(
      elevation: 0,
      shape: RoundedRectangleBorder(borderRadius:
        BorderRadius.circular(16)),
      color: darkSurface,
    ),
    appBarTheme: const AppBarTheme(
      elevation: 0,
      centerTitle: true,
      backgroundColor: darkSurface,
      foregroundColor: textOnDark,
    ),
    bottomNavigationBarTheme: const BottomNavigationBarThemeData(
      backgroundColor: darkSurface,
      selectedItemColor: primaryColor,
      unselectedItemColor: textSecondary,
      type: BottomNavigationBarType.fixed,
    ),
  );
}
```

5.2 Connection Screen (Main)

```
// lib/ui/screens/connection_screen.dart
import 'package:flutter/material.dart';
import 'package:flutter_riverpod/flutter_riverpod.dart';

class ConnectionScreen extends ConsumerWidget {
  const ConnectionScreen({super.key});

  @override
  Widget build(BuildContext context, WidgetRef ref) {
```

```
  final connectionState = ref.watch(connectionStateProvider);
  final selectedServer = ref.watch(selectedServerProvider);
  final connectionStats = ref.watch(connectionStatsProvider);
  final isDarkMode = ref.watch(themeProvider);

  return Scaffold(
    body: SafeArea(
      child: Column(
        children: [
          // Header with current server
          _buildServerSelector(context, selectedServer),

          const Spacer(),

          // Connection status and toggle
          _buildConnectionStatus(connectionState),
          const SizedBox(height: 32),

          // Main connection button
          ConnectionButton(
            state: connectionState,
            onToggle: () => _toggleConnection(ref,
      connectionState),
          ),

          const Spacer(),

          // Connection stats (visible when connected)
          if (connectionState == ConnectionState.connected)
            ConnectionStatsPanel(stats: connectionStats),

          const SizedBox(height: 24),
        ],
      ),
    ),
  );
}

Widget _buildServerSelector(BuildContext context, Server?
      server) {
  return Padding(
    padding: const EdgeInsets.all(16),
    child: InkWell(
      onTap: () => Navigator.pushNamed(context, '/servers'),
      borderRadius: BorderRadius.circular(12),
      child: Container(
        padding: const EdgeInsets.symmetric(horizontal: 16,
      vertical: 12),
        decoration: BoxDecoration(
          borderRadius: BorderRadius.circular(12),
          color: Theme.of(context).cardTheme.color,
        ),
```

```
        child: Row(
          children: [
            // Flag icon
            CountryFlag(
              countryCode: server?.countryCode ?? 'auto',
              size: 32,
            ),
            const SizedBox(width: 12),

            // Server info
            Expanded(
              child: Column(
                crossAxisAlignment: CrossAxisAlignment.start,
                children: [
                  Text(
                    server?.displayName ?? 'Optimal Location',
                    style: const TextStyle(
                      fontSize: 16,
                      fontWeight: FontWeight.w600,
                    ),
                  ),
                  if (server != null)
                    Text(
                      '${server.city} \u00B7 ${server.load}%
      load',
                      style: TextStyle(
                        fontSize: 12,
                        color:
      Theme.of(context).colorScheme.onSurface.withOpacity(0.6),
                      ),
                    ),
                ],
              ),
            ),

            // Latency indicator
            if (server != null)
              _buildLatencyIndicator(server.latencyMs),

            const Icon(Icons.chevron_right),
          ],
        ),
      ),
    ),
  );
}

Widget _buildConnectionStatus(ConnectionState state) {
  String statusText;
  Color statusColor;

  switch (state) {
```

```
      case ConnectionState.connected:
        statusText = 'Connected';
        statusColor = HelixTheme.connected;
      case ConnectionState.connecting:
        statusText = 'Connecting...';
        statusColor = HelixTheme.connecting;
      case ConnectionState.disconnecting:
        statusText = 'Disconnecting...';
        statusColor = HelixTheme.warningColor;
      case ConnectionState.disconnected:
        statusText = 'Not Connected';
        statusColor = HelixTheme.disconnected;
      case ConnectionState.error:
        statusText = 'Connection Error';
        statusColor = HelixTheme.error;
    }

    return Column(
      children: [
        // Status dot
        Container(
          width: 8,
          height: 8,
          decoration: BoxDecoration(
            shape: BoxShape.circle,
            color: statusColor,
            boxShadow: [
              BoxShadow(
                color: statusColor.withOpacity(0.4),
                blurRadius: 8,
                spreadRadius: 2,
              ),
            ],
          ),
        ),
        const SizedBox(height: 8),
        Text(
          statusText,
          style: TextStyle(
            fontSize: 18,
            fontWeight: FontWeight.w600,
            color: statusColor,
          ),
        ),
      ],
    );
  }

  Widget _buildLatencyIndicator(int latencyMs) {
    Color color;
    if (latencyMs < 50) color = HelixTheme.connected;
    else if (latencyMs < 100) color = HelixTheme.warningColor;
```

```
    else color = HelixTheme.error;

    return Container(
      padding: const EdgeInsets.symmetric(horizontal: 8,
        vertical: 4),
      decoration: BoxDecoration(
        color: color.withOpacity(0.1),
        borderRadius: BorderRadius.circular(8),
      ),
      child: Text(
        '${latencyMs}ms',
        style: TextStyle(
          fontSize: 12,
          color: color,
          fontWeight: FontWeight.w500,
        ),
      ),
    );
  }

  void _toggleConnection(WidgetRef ref, ConnectionState state) {
    switch (state) {
      case ConnectionState.disconnected:
      case ConnectionState.error:
        ref.read(connectionStateProvider.notifier).connect();
      case ConnectionState.connected:
        ref.read(connectionStateProvider.notifier).disconnect();
      default:
        // Do nothing while transitioning
        break;
    }
  }
}
```

5.3 Connection Button with Animation

```
// lib/ui/widgets/connection_button.dart
import 'package:flutter/material.dart';

class ConnectionButton extends StatefulWidget {
  final ConnectionState state;
  final VoidCallback onToggle;

  const ConnectionButton({
    super.key,
    required this.state,
    required this.onToggle,
  });

  @override
```

```
  State<ConnectionButton> createState() =>
        _ConnectionButtonState();
}

class _ConnectionButtonState extends State<ConnectionButton>
    with TickerProviderStateMixin {
  late AnimationController _pulseController;
  late AnimationController _rotateController;
  late Animation<double> _pulseAnimation;

  @override
  void initState() {
    super.initState();

    _pulseController = AnimationController(
      vsync: this,
      duration: const Duration(milliseconds: 1500),
    )..repeat(reverse: true);

    _rotateController = AnimationController(
      vsync: this,
      duration: const Duration(milliseconds: 2000),
    );

    _pulseAnimation = Tween<double>(begin: 1.0, end:
        1.15).animate(
      CurvedAnimation(
        parent: _pulseController,
        curve: Curves.easeInOut,
      ),
    );
  }

  @override
  void didUpdateWidget(ConnectionButton oldWidget) {
    super.didUpdateWidget(oldWidget);

    if (widget.state == ConnectionState.connecting) {
      _rotateController.repeat();
    } else {
      _rotateController.stop();
    }

    if (widget.state == ConnectionState.connected) {
      _pulseController.repeat(reverse: true);
    } else if (widget.state != ConnectionState.connecting) {
      _pulseController.stop();
      _pulseController.value = 0;
    }
  }

  @override
```

```
void dispose() {
  _pulseController.dispose();
  _rotateController.dispose();
  super.dispose();
}

@override
Widget build(BuildContext context) {
  final isConnected = widget.state == ConnectionState.connected;
  final isConnecting = widget.state ==
      ConnectionState.connecting;
  final buttonColor = isConnected ? HelixTheme.connected :
      HelixTheme.disconnected;

  return AnimatedBuilder(
    animation: Listenable.merge([_pulseController,
      _rotateController]),
    builder: (context, child) {
      return Transform.scale(
        scale: isConnected ? _pulseAnimation.value : 1.0,
        child: GestureDetector(
          onTap: widget.onToggle,
          child: Container(
            width: 140,
            height: 140,
            decoration: BoxDecoration(
              shape: BoxShape.circle,
              gradient: RadialGradient(
                colors: [
                  buttonColor.withOpacity(0.3),
                  buttonColor.withOpacity(0.1),
                  Colors.transparent,
                ],
                stops: const [0.0, 0.6, 1.0],
              ),
            ),
            child: Center(
              child: Container(
                width: 100,
                height: 100,
                decoration: BoxDecoration(
                  shape: BoxShape.circle,
                  gradient: LinearGradient(
                    begin: Alignment.topLeft,
                    end: Alignment.bottomRight,
                    colors: [
                      buttonColor,
                      buttonColor.withOpacity(0.8),
                    ],
                  ),
                  boxShadow: [
                    BoxShadow(
```

```
                        color: buttonColor.withOpacity(0.4),
                        blurRadius: 30,
                        spreadRadius: isConnected ? 10 : 0,
                      ),
                    ],
                  ),
                  child: Center(
                    child: isConnecting
                        ? RotationTransition(
                            turns: _rotateController,
                            child: const Icon(
                              Icons.sync,
                              color: Colors.white,
                              size: 40,
                            ),
                          )
                        : Icon(
                            isConnected ?
        Icons.power_settings_new : Icons.power_settings_new,
                            color: Colors.white,
                            size: 48,
                          ),
                  ),
                ),
              ),
            ),
          ),
        );
      },
    );
  }
}

enum ConnectionState {
  disconnected,
  connecting,
  connected,
  disconnecting,
  error,
}
```

5.4 Server Selection Screen

```
// lib/ui/screens/server_selection_screen.dart
import 'package:flutter/material.dart';
import 'package:flutter_riverpod/flutter_riverpod.dart';

class ServerSelectionScreen extends ConsumerStatefulWidget {
  const ServerSelectionScreen({super.key});

  @override
```

```
  ConsumerState<ServerSelectionScreen> createState() =>
      _ServerSelectionScreenState();
}

class _ServerSelectionScreenState extends
      ConsumerState<ServerSelectionScreen> {
  String _searchQuery = '';
  ServerSortMode _sortMode = ServerSortMode.latency;

  @override
  Widget build(BuildContext context) {
    final servers = ref.watch(serverListProvider);
    final selectedServer = ref.watch(selectedServerProvider);
    final favorites = ref.watch(favoriteServersProvider);

    final filteredServers = servers.where((server) {
      final matchesSearch = server.displayName.toLowerCase()
          .contains(_searchQuery.toLowerCase()) ||

          server.city.toLowerCase().contains(_searchQuery.toLowerCase())
          ||

          server.country.toLowerCase().contains(_searchQuery.toLowerCase());
      return matchesSearch;
    }).toList();

    // Sort servers
    filteredServers.sort((a, b) {
      switch (_sortMode) {
        case ServerSortMode.latency:
          return a.latencyMs.compareTo(b.latencyMs);
        case ServerSortMode.load:
          return a.load.compareTo(b.load);
        case ServerSortMode.name:
          return a.displayName.compareTo(b.displayName);
      }
    });

    return Scaffold(
      appBar: AppBar(
        title: const Text('Select Server'),
        actions: [
          PopupMenuButton<ServerSortMode>(
            icon: const Icon(Icons.sort),
            onSelected: (mode) => setState(() => _sortMode =
      mode),
              itemBuilder: (context) => [
                const PopupMenuItem(
                  value: ServerSortMode.latency,
                  child: Text('Sort by Latency'),
                ),
                const PopupMenuItem(
                  value: ServerSortMode.load,
```

```
            child: Text('Sort by Load'),
          ),
          const PopupMenuItem(
            value: ServerSortMode.name,
            child: Text('Sort by Name'),
          ),
        ],
      ),
    ],
  ),
  body: Column(
    children: [
      // Search bar
      Padding(
        padding: const EdgeInsets.all(16),
        child: TextField(
          decoration: InputDecoration(
            hintText: 'Search servers...',
            prefixIcon: const Icon(Icons.search),
            border: OutlineInputBorder(
              borderRadius: BorderRadius.circular(12),
            ),
            filled: true,
          ),
          onChanged: (value) => setState(() => _searchQuery =
    value),
        ),
      ),

      // Quick connect button
      Padding(
        padding: const EdgeInsets.symmetric(horizontal: 16),
        child: ListTile(
          leading: const Icon(Icons.bolt, color:
    HelixTheme.accentColor),
          title: const Text('Optimal Location'),
          subtitle: const Text('Best performance
    automatically'),
          trailing: selectedServer == null
              ? const Icon(Icons.check_circle, color:
    HelixTheme.accentColor)
              : null,
          onTap: () {

    ref.read(selectedServerProvider.notifier).selectOptimal();
            Navigator.pop(context);
          },
        ),
      ),

      const Divider(),

      // Server list
```

```
          Expanded(
            child: ListView.builder(
              itemCount: filteredServers.length,
              itemBuilder: (context, index) {
                final server = filteredServers[index];
                final isSelected = server.id ==
        selectedServer?.id;
                final isFavorite = favorites.contains(server.id);

                return ServerListTile(
                  server: server,
                  isSelected: isSelected,
                  isFavorite: isFavorite,
                  onTap: () {

        ref.read(selectedServerProvider.notifier).select(server);
                    Navigator.pop(context);
                  },
                  onFavoriteToggle: () {
                    ref.read(favoriteServersProvider.notifier)
                        .toggle(server.id);
                  },
                );
              },
            ),
          ),
        ],
      ),
    );
  }
}

class ServerListTile extends StatelessWidget {
  final Server server;
  final bool isSelected;
  final bool isFavorite;
  final VoidCallback onTap;
  final VoidCallback onFavoriteToggle;

  const ServerListTile({
    super.key,
    required this.server,
    required this.isSelected,
    required this.isFavorite,
    required this.onTap,
    required this.onFavoriteToggle,
  });

  @override
  Widget build(BuildContext context) {
    Color latencyColor;
```

```
if (server.latencyMs < 50) latencyColor =
    HelixTheme.connected;
else if (server.latencyMs < 100) latencyColor =
    HelixTheme.warningColor;
else latencyColor = HelixTheme.error;

return ListTile(
  leading: Stack(
    children: [
      CountryFlag(countryCode: server.countryCode, size: 36),
      if (isSelected)
        Positioned(
          right: 0,
          bottom: 0,
          child: Container(
            padding: const EdgeInsets.all(2),
            decoration: const BoxDecoration(
              color: HelixTheme.connected,
              shape: BoxShape.circle,
            ),
            child: const Icon(Icons.check, size: 10, color:
    Colors.white),
          ),
        ),
    ],
  ),
  title: Text(server.displayName),
  subtitle: Row(
    children: [
      Container(
        width: 6,
        height: 6,
        decoration: BoxDecoration(
          shape: BoxShape.circle,
          color: latencyColor,
        ),
      ),
      const SizedBox(width: 4),
      Text('${server.latencyMs}ms'),
      const SizedBox(width: 12),
      Text('${server.load}% load'),
    ],
  ),
  trailing: Row(
    mainAxisSize: MainAxisSize.min,
    children: [
      // Protocol support badges
      if (server.supportsWireGuard)
        _buildProtocolBadge('WG'),
      if (server.supportsOpenVPN)
        _buildProtocolBadge('OV'),
      const SizedBox(width: 8),
```

```
          // Favorite toggle
          IconButton(
            icon: Icon(
              isFavorite ? Icons.star : Icons.star_border,
              color: isFavorite ? HelixTheme.warningColor : null,
            ),
            onPressed: onFavoriteToggle,
          ),
        ],
      ),
      onTap: onTap,
      selected: isSelected,
    );
  }

  Widget _buildProtocolBadge(String label) {
    return Container(
      margin: const EdgeInsets.only(right: 4),
      padding: const EdgeInsets.symmetric(horizontal: 4,
        vertical: 2),
      decoration: BoxDecoration(
        color: HelixTheme.primaryColor.withOpacity(0.1),
        borderRadius: BorderRadius.circular(4),
      ),
      child: Text(
        label,
        style: const TextStyle(
          fontSize: 10,
          color: HelixTheme.primaryColor,
          fontWeight: FontWeight.w500,
        ),
      ),
    );
  }
}

enum ServerSortMode { latency, load, name }
```

5.5 Settings Screen

```
// lib/ui/screens/settings_screen.dart
class SettingsScreen extends ConsumerWidget {
  const SettingsScreen({super.key});

  @override
  Widget build(BuildContext context, WidgetRef ref) {
    final settings = ref.watch(appSettingsProvider);

    return Scaffold(
      appBar: AppBar(title: const Text('Settings')),
```

```
      body: ListView(
        children: [
          // Connection Settings
          _buildSectionHeader(context, 'Connection'),

          ListTile(
            leading: const Icon(Icons.security),
            title: const Text('Protocol'),
            subtitle: Text(settings.protocol.displayName),
            trailing: const Icon(Icons.chevron_right),
            onTap: () => _showProtocolSelector(context, ref),
          ),

          ListTile(
            leading: const Icon(Icons.call_split),
            title: const Text('Split Tunneling'),
            subtitle: Text(settings.splitTunnelEnabled ?
        'Enabled' : 'Disabled'),
            trailing: const Icon(Icons.chevron_right),
            onTap: () => Navigator.pushNamed(context, '/
        split_tunneling'),
          ),

          ListTile(
            leading: const Icon(Icons.shield),
            title: const Text('Kill Switch'),
            subtitle: Text(settings.killSwitchEnabled ?
        'Enabled' : 'Disabled'),
            trailing: Switch(
              value: settings.killSwitchEnabled,
              onChanged: (value) {

        ref.read(appSettingsProvider.notifier).setKillSwitch(value);
              },
            ),
          ),

          // Auto-Connect Settings
          _buildSectionHeader(context, 'Auto-Connect'),

          ListTile(
            leading: const Icon(Icons.wifi),
            title: const Text('Connect on Untrusted Wi-Fi'),
            trailing: Switch(
              value: settings.connectOnUntrustedWifi,
              onChanged: (value) {

        ref.read(appSettingsProvider.notifier).setConnectOnUntrustedWifi(value);
              },
            ),
          ),

          ListTile(
```

```
      leading: const Icon(Icons.signal_cellular_alt),
      title: const Text('Connect on Cellular'),
      trailing: Switch(
        value: settings.connectOnCellular,
        onChanged: (value) {
ref.read(appSettingsProvider.notifier).setConnectOnCellular(value);
        },
      ),
    ),

    ListTile(
      leading: const Icon(Icons.network_check),
      title: const Text('Trusted Wi-Fi Networks'),
      subtitle:
Text('${settings.trustedWifiNetworks.length} networks'),
      trailing: const Icon(Icons.chevron_right),
      onTap: () => Navigator.pushNamed(context, '/
trusted_wifi'),
    ),

    // Security Settings
    _buildSectionHeader(context, 'Security'),

    ListTile(
      leading: const Icon(Icons.fingerprint),
      title: const Text('Biometric Authentication'),
      subtitle: Text(settings.biometricAuth ? 'Enabled' :
'Disabled'),
      trailing: Switch(
        value: settings.biometricAuth,
        onChanged: (value) => _toggleBiometric(context,
ref, value),
      ),
    ),

    ListTile(
      leading: const Icon(Icons.dns),
      title: const Text('Custom DNS'),
      subtitle: Text(settings.customDns?.join(', ') ??
'Default'),
      trailing: const Icon(Icons.chevron_right),
      onTap: () => _showDnsConfig(context, ref),
    ),

    // App Settings
    _buildSectionHeader(context, 'App'),

    ListTile(
      leading: const Icon(Icons.dark_mode),
      title: const Text('Dark Mode'),
      trailing: DropdownButton<ThemeMode>(
        value: settings.themeMode,
```

```
              underline: const SizedBox.shrink(),
              items: ThemeMode.values.map((mode) {
                return DropdownMenuItem(
                  value: mode,
                  child: Text(mode.displayName),
                );
              }).toList(),
              onChanged: (mode) {
                if (mode != null) {

        ref.read(appSettingsProvider.notifier).setThemeMode(mode);
                }
              },
            ),
          ),

          ListTile(
            leading: const Icon(Icons.vibration),
            title: const Text('Haptic Feedback'),
            trailing: Switch(
              value: settings.hapticFeedback,
              onChanged: (value) {

        ref.read(appSettingsProvider.notifier).setHapticFeedback(value);
              },
            ),
          ),

          const AboutListTile(
            icon: Icon(Icons.info),
            applicationName: 'Helix VPN',
            applicationVersion: '1.0.0',
            applicationLegalese: '\u00A9 2025 Helix VPN Inc.',
          ),
        ],
      ),
    );
  }

  Widget _buildSectionHeader(BuildContext context, String title) {
    return Padding(
      padding: const EdgeInsets.fromLTRB(16, 24, 16, 8),
      child: Text(
        title.toUpperCase(),
        style: TextStyle(
          fontSize: 12,
          fontWeight: FontWeight.w600,
          color: Theme.of(context).colorScheme.primary,
          letterSpacing: 1,
        ),
      ),
    );
```

```
  }
}
```

5.6 Split Tunneling Configuration Screen

```dart
// lib/ui/screens/split_tunneling_screen.dart
class SplitTunnelingScreen extends ConsumerStatefulWidget {
  const SplitTunnelingScreen({super.key});

  @override
  ConsumerState<SplitTunnelingScreen> createState() =>
        _SplitTunnelingScreenState();
}

class _SplitTunnelingScreenState extends
        ConsumerState<SplitTunnelingScreen>
    with SingleTickerProviderStateMixin {
  late TabController _tabController;
  SplitTunnelMode _mode = SplitTunnelMode.off;
  final Set<String> _selectedApps = {};
  List<InstalledApp> _installedApps = [];
  bool _isLoading = true;

  @override
  void initState() {
    super.initState();
    _tabController = TabController(length: 3, vsync: this);
    _loadApps();
  }

  Future<void> _loadApps() async {
    final apps = await
        ref.read(vpnServiceProvider).getInstalledApps();
    setState(() {
      _installedApps = apps;
      _isLoading = false;
    });
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('Split Tunneling'),
        bottom: TabBar(
          controller: _tabController,
          tabs: const [
            Tab(text: 'Off'),
            Tab(text: 'Include'),
            Tab(text: 'Exclude'),
          ],
          onTap: (index) {
```

```
        setState(() {
          _mode = SplitTunnelMode.values[index];
        });
      },
    ),
  ),
),
body: _isLoading
    ? const Center(child: CircularProgressIndicator())
    : Column(
        children: [
          // Mode explanation
          Padding(
            padding: const EdgeInsets.all(16),
            child: _buildModeExplanation(),
          ),

          // App selection list
          Expanded(
            child: ListView.builder(
              itemCount: _installedApps.length,
              itemBuilder: (context, index) {
                final app = _installedApps[index];
                final isSelected =
  _selectedApps.contains(app.packageName);

                return CheckboxListTile(
                  secondary: app.icon != null
                      ? Image.memory(app.icon!, width: 40,
  height: 40)
                      : const Icon(Icons.android, size: 40),
                  title: Text(app.name),
                  subtitle: Text(app.packageName),
                  value: isSelected,
                  onChanged: _mode == SplitTunnelMode.off
                      ? null
                      : (value) {
                          setState(() {
                            if (value == true) {

  _selectedApps.add(app.packageName);
                            } else {

  _selectedApps.remove(app.packageName);
                            }
                          });
                        },
                );
              },
            ),
          ),

          // Save button
```

```
              Padding(
                padding: const EdgeInsets.all(16),
                child: SizedBox(
                  width: double.infinity,
                  child: FilledButton(
                    onPressed: () => _saveConfiguration(),
                    child: const Text('Save Configuration'),
                  ),
                ),
              ),
            ],
          ),
    );
}

Widget _buildModeExplanation() {
  String explanation;
  switch (_mode) {
    case SplitTunnelMode.off:
      explanation = 'All app traffic will be routed through the
      VPN.';
    case SplitTunnelMode.include:
      explanation = 'Only selected apps will use the VPN. All
      other apps will connect directly.';
    case SplitTunnelMode.exclude:
      explanation = 'Selected apps will bypass the VPN. All
      other apps will use the VPN.';
  }

  return Container(
    padding: const EdgeInsets.all(12),
    decoration: BoxDecoration(
      color:
      Theme.of(context).colorScheme.primary.withOpacity(0.1),
      borderRadius: BorderRadius.circular(8),
    ),
    child: Row(
      children: [
        Icon(Icons.info_outline,
            color: Theme.of(context).colorScheme.primary),
        const SizedBox(width: 12),
        Expanded(child: Text(explanation)),
      ],
    ),
  );
}

void _saveConfiguration() {
  ref.read(vpnServiceProvider).setSplitTunneling(
    mode: _mode,
    apps: _selectedApps.toList(),
  );
```

```
    ScaffoldMessenger.of(context).showSnackBar(
      const SnackBar(content: Text('Split tunneling configuration
        saved')),
    );
  }
}

enum SplitTunnelMode { off, include, exclude }
```

5.7 Notification Design

```
// lib/services/notification_service.dart
import 'package:flutter_local_notifications/
        flutter_local_notifications.dart';

class NotificationService {
  static final NotificationService _instance =
        NotificationService._internal();
  factory NotificationService() => _instance;
  NotificationService._internal();

  final FlutterLocalNotificationsPlugin _notifications =
      FlutterLocalNotificationsPlugin();

  Future<void> initialize() async {
    const androidSettings =
        AndroidInitializationSettings('@drawable/
        ic_notification');
    const iosSettings = DarwinInitializationSettings(
      requestAlertPermission: false,
      requestBadgePermission: false,
      requestSoundPermission: false,
    );

    const initSettings = InitializationSettings(
      android: androidSettings,
      iOS: iosSettings,
    );

    await _notifications.initialize(initSettings);
  }

  // Android: Foreground service notification (handled natively)
  // This is for Flutter-triggered notifications only

  Future<void> showConnectionNotification({
    required String serverLocation,
    required String protocol,
  }) async {
    const androidDetails = AndroidNotificationDetails(
      'vpn_status',
      'VPN Connection Status',
```

```
      channelDescription: 'Shows when Helix VPN is connected',
      importance: Importance.low,
      priority: Priority.low,
      ongoing: true,
      autoCancel: false,
      showWhen: false,
    );

    const iosDetails = DarwinNotificationDetails(
      presentAlert: false,
      presentBadge: false,
      presentSound: false,
    );

    const details = NotificationDetails(
      android: androidDetails,
      iOS: iosDetails,
    );

    await _notifications.show(
      1,
      'Helix VPN Connected',
      'Server: $serverLocation \u00B7 $protocol',
      details,
    );
  }

  Future<void> showReconnectionWarning() async {
    const androidDetails = AndroidNotificationDetails(
      'vpn_alerts',
      'VPN Alerts',
      channelDescription: 'Important VPN connection alerts',
      importance: Importance.high,
      priority: Priority.high,
    );

    await _notifications.show(
      2,
      'Connection Unstable',
      'Helix VPN is reconnecting to maintain your secure
        connection.',
      const NotificationDetails(android: androidDetails),
    );
  }

  Future<void> cancelAll() async {
    await _notifications.cancelAll();
  }
}
```

5.8 Widget Support

Android Home Screen Widget:

```
<!-- android/app/src/main/res/xml/vpn_widget_info.xml -->
<appwidget-provider xmlns:android="http://schemas.android.com/apk/
        res/android"
    android:minWidth="180dp"
    android:minHeight="40dp"
    android:updatePeriodMillis="0"
    android:initialLayout="@layout/vpn_widget"
    android:previewImage="@drawable/widget_preview"
    android:widgetCategory="home_screen"
    android:resizeMode="horizontal" />

// VpnAppWidgetProvider.kt
class VpnAppWidgetProvider : AppWidgetProvider() {
    override fun onUpdate(
        context: Context,
        appWidgetManager: AppWidgetManager,
        appWidgetIds: IntArray
    ) {
        for (appWidgetId in appWidgetIds) {
            updateWidget(context, appWidgetManager, appWidgetId)
        }
    }

    private fun updateWidget(
        context: Context,
        appWidgetManager: AppWidgetManager,
        appWidgetId: Int
    ) {
        val views = RemoteViews(context.packageName,
        R.layout.vpn_widget)

        val isConnected = HelixVpnService.isRunning

        views.setTextViewText(
            R.id.widget_status,
            if (isConnected) "Connected" else "Disconnected"
        )
        views.setImageViewResource(
            R.id.widget_icon,
            if (isConnected) R.drawable.ic_widget_on else
        R.drawable.ic_widget_off
        )

        // Toggle intent
        val toggleIntent = PendingIntent.getBroadcast(
            context,
            0,
```

```
            Intent(context,
        VpnAppWidgetProvider::class.java).apply {
                action = "TOGGLE_VPN"
            },
            PendingIntent.FLAG_UPDATE_CURRENT or
        PendingIntent.FLAG_IMMUTABLE
        )
        views.setOnClickPendingIntent(R.id.widget_button,
        toggleIntent)

        appWidgetManager.updateAppWidget(appWidgetId, views)
    }

    override fun onReceive(context: Context, intent: Intent) {
        super.onReceive(context, intent)

        if (intent.action == "TOGGLE_VPN") {
            if (HelixVpnService.isRunning) {
                context.startService(
                    Intent(context, HelixVpnService::class.java)
                        .setAction(HelixVpnService.ACTION_DISCONNECT)
                )
            } else {
                // Launch app for connection
                context.startActivity(

        context.packageManager.getLaunchIntentForPackage(context.packageName)
                )
            }
        }
    }
}
```

iOS Widget (using SwiftUI):

```
// HelixVpnWidget/HelixVpnWidget.swift
import WidgetKit
import SwiftUI

struct HelixVpnWidgetEntry: TimelineEntry {
    let date: Date
    let isConnected: Bool
    let serverLocation: String
}

struct HelixVpnWidgetProvider: TimelineProvider {
    func placeholder(in context: Context) -> HelixVpnWidgetEntry {
        HelixVpnWidgetEntry(date: Date(), isConnected: false,
        serverLocation: "Not Connected")
    }

    func getSnapshot(in context: Context, completion: @escaping
        (HelixVpnWidgetEntry) -> Void) {
```

```
        let entry = loadCurrentState()
        completion(entry)
    }

    func getTimeline(in context: Context, completion: @escaping
        (Timeline<HelixVpnWidgetEntry>) -> Void) {
        let entry = loadCurrentState()
        let timeline = Timeline(entries: [entry],
        policy: .after(Date().addingTimeInterval(60)))
        completion(timeline)
    }

    private func loadCurrentState() -> HelixVpnWidgetEntry {
        let sharedDefaults = UserDefaults(suiteName:
        "group.com.helix.vpn")
        let isConnected = sharedDefaults?.string(forKey:
        "tunnel_state") == "connected"
        let serverLocation = sharedDefaults?.string(forKey:
        "last_server") ?? "Not Connected"

        return HelixVpnWidgetEntry(
            date: Date(),
            isConnected: isConnected,
            serverLocation: serverLocation
        )
    }
}

struct HelixVpnWidgetView: View {
    var entry: HelixVpnWidgetProvider.Entry
    @Environment(\.widgetFamily) var family

    var body: some View {
        VStack(spacing: 8) {
            Image(systemName: entry.isConnected ?
        "shield.checkered" : "shield")
                .font(.title2)
                .foregroundColor(entry.isConnected ? .green : .gray)

            Text(entry.isConnected ? "Connected" : "Disconnected")
                .font(.caption)
                .fontWeight(.semibold)

            if entry.isConnected {
                Text(entry.serverLocation)
                    .font(.caption2)
                    .foregroundColor(.secondary)
                    .lineLimit(1)
            }
        }
        .padding()
        .containerBackground(.fill.tertiary, for: .widget)
    }
```

```
}

@main
struct HelixVpnWidget: Widget {
    let kind: String = "HelixVpnWidget"

    var body: some WidgetConfiguration {
        StaticConfiguration(kind: kind, provider:
        HelixVpnWidgetProvider()) { entry in
            HelixVpnWidgetView(entry: entry)
        }
        .configurationDisplayName("Helix VPN Status")
        .description("Quickly see your VPN connection status.")
        .supportedFamilies([.systemSmall, .systemMedium])
    }
}
```

---

6. Mobile-Specific Features

6.1 Biometric Authentication

```dart
// lib/services/biometric_service.dart
import 'package:local_auth/local_auth.dart';
import 'package:local_auth_android/local_auth_android.dart';
import 'package:local_auth_darwin/local_auth_darwin.dart';

class BiometricService {
  static final LocalAuthentication _localAuth =
        LocalAuthentication();

  /// Check if biometric authentication is available
  static Future<bool> isAvailable() async {
    final isDeviceSupported = await
        _localAuth.isDeviceSupported();
    final canCheck = await _localAuth.canCheckBiometrics;
    return isDeviceSupported && canCheck;
  }

  /// Get available biometric types
  static Future<List<BiometricType>> getAvailableTypes() async {
    return await _localAuth.getAvailableBiometrics();
  }

  /// Authenticate user with biometrics
  static Future<bool> authenticate({
    String localizedReason = 'Authenticate to access VPN',
    bool useErrorDialogs = true,
    bool stickyAuth = false,
    bool sensitiveTransaction = true,
```

```
}) async {
  try {
    return await _localAuth.authenticate(
      localizedReason: localizedReason,
      authMessages: const [
        AndroidAuthMessages(
          signInTitle: 'Helix VPN Authentication',
          cancelButton: 'Cancel',
          biometricHint: 'Verify your identity',
          biometricNotRecognized:
      'Biometric not recognized, try again',
          biometricRequiredTitle: 'Biometric authentication
      required',
          deviceCredentialsRequiredTitle: 'Device credentials
      required',
          deviceCredentialsSetupDescription: 'Please set up
      device credentials',
          goToSettingsButton: 'Go to Settings',
          goToSettingsDescription: 'Please set up biometric
      authentication in Settings',
        ),
        IOSAuthMessages(
          cancelButton: 'Cancel',
          goToSettingsButton: 'Go to Settings',
          goToSettingsDescription: 'Please set up biometric
      authentication in Settings',
          lockOut: 'Biometric authentication is locked out',
        ),
      ],
      options: AuthenticationOptions(
        useErrorDialogs: useErrorDialogs,
        stickyAuth: stickyAuth,
        sensitiveTransaction: sensitiveTransaction,
        biometricOnly: false, // Allow fallback to PIN/password
      ),
    );
  } catch (e) {
    return false;
  }
}

/// Authenticate before VPN connection
static Future<bool> authenticateForConnection() async {
  return authenticate(
    localizedReason: 'Authenticate to connect Helix VPN',
    sensitiveTransaction: true,
  );
}

/// Authenticate before accessing sensitive settings
static Future<bool> authenticateForSettings() async {
  return authenticate(
    localizedReason: 'Authenticate to access VPN settings',
```

```
      sensitiveTransaction: true,
    );
  }
}
```

6.2 Auto-Connect on Untrusted Wi-Fi

```dart
// lib/services/auto_connect_service.dart
import 'package:connectivity_plus/connectivity_plus.dart';
import 'package:wifi_info_flutter/wifi_info_flutter.dart';

class AutoConnectService {
  final Ref ref;
  final Connectivity _connectivity = Connectivity();
  StreamSubscription? _connectivitySubscription;

  AutoConnectService(this.ref);

  void startMonitoring() {
    _connectivitySubscription =
        _connectivity.onConnectivityChanged
        .listen(_handleConnectivityChange);
  }

  void stopMonitoring() {
    _connectivitySubscription?.cancel();
  }

  Future<void> _handleConnectivityChange(ConnectivityResult
        result) async {
    final settings = ref.read(appSettingsProvider);
    if (!settings.autoConnectEnabled) return;

    if (result == ConnectivityResult.wifi) {
      final wifiName = await WifiInfo().getWifiName();
      final isTrusted =
          settings.trustedWifiNetworks.contains(wifiName);

      if (!isTrusted && settings.connectOnUntrustedWifi) {
        // Auto-connect on untrusted Wi-Fi
        await
          ref.read(connectionStateProvider.notifier).connect();
      } else if (isTrusted && settings.disconnectOnTrustedWifi) {
        // Disconnect on trusted Wi-Fi
        await
          ref.read(connectionStateProvider.notifier).disconnect();
      }
    } else if (result == ConnectivityResult.mobile) {
      if (settings.connectOnCellular) {
        await
          ref.read(connectionStateProvider.notifier).connect();
      }
```

```
    }
  }
}
```

6.3 Siri Shortcuts (iOS) / App Actions (Android)

iOS Siri Shortcuts:

```swift
// Runner/AppDelegate.swift additions
import Intents

// Register Siri intents for VPN control
class IntentHandler: INExtension {
    override func handler(for intent: INIntent) -> Any {
        if intent is ConnectVpnIntent {
            return ConnectVpnIntentHandler()
        } else if intent is DisconnectVpnIntent {
            return DisconnectVpnIntentHandler()
        }
        return self
    }
}

class ConnectVpnIntentHandler: NSObject, ConnectVpnIntentHandling
        {
    func handle(intent: ConnectVpnIntent, completion: @escaping
        (ConnectVpnIntentResponse) -> Void) {
        // Trigger VPN connection
        VpnConfigurationManager.shared.loadOrCreateConfiguration
        { manager, error in
            guard let manager = manager else {

        completion(ConnectVpnIntentResponse(code: .failure,
        userActivity: nil))
                return
            }
            VpnConfigurationManager.shared.connect(manager:
        manager, config: defaultConfig) { error in
                let response = ConnectVpnIntentResponse(
                    code: error == nil ? .success : .failure,
                    userActivity: nil
                )
                completion(response)
            }
        }
    }
}
```

Android App Actions:

```xml
<!-- android/app/src/main/res/xml/actions.xml -->
<actions>
```

```
    <action intentName="actions.intent.OPEN_APP_FEATURE">
        <fulfillment
            fulfillmentMode="actions.fulfillment.DEEPLINK"
            urlTemplate="helixvpn://connect" />
        <fulfillment
            fulfillmentMode="actions.fulfillment.DEEPLINK"
            urlTemplate="helixvpn://disconnect" />
        <parameter name="feature">
            <entity-set-reference
         entitySetId="VpnFeatureEntitySet" />
        </parameter>
    </action>
</actions>
```

6.4 Connection Stats and Data Usage

```
// lib/models/connection_stats.dart
import 'package:freezed_annotation/freezed_annotation.dart';

part 'connection_stats.freezed.dart';
part 'connection_stats.g.dart';

@freezed
class ConnectionStats with _$ConnectionStats {
  const factory ConnectionStats({
    @Default(0) double uploadSpeed,
    @Default(0) double downloadSpeed,
    @Default(0) int totalUpload,
    @Default(0) int totalDownload,
    @Default(0) int connectionDuration,
    String? serverLocation,
    String? protocol,
    @Default(0) int pingMs,
    @Default([]) List<DataUsagePoint> usageHistory,
  }) = _ConnectionStats;

  factory ConnectionStats.fromJson(Map<String, dynamic> json) =>
      _$ConnectionStatsFromJson(json);
}

@freezed
class DataUsagePoint with _$DataUsagePoint {
  const factory DataUsagePoint({
    required DateTime timestamp,
    required double uploadSpeed,
    required double downloadSpeed,
  }) = _DataUsagePoint;

  factory DataUsagePoint.fromJson(Map<String, dynamic> json) =>
```

```
      _$DataUsagePointFromJson(json);
}

// lib/ui/widgets/connection_stats_panel.dart
class ConnectionStatsPanel extends StatelessWidget {
  final ConnectionStats stats;

  const ConnectionStatsPanel({super.key, required this.stats});

  @override
  Widget build(BuildContext context) {
    return Container(
      margin: const EdgeInsets.symmetric(horizontal: 16),
      padding: const EdgeInsets.all(16),
      decoration: BoxDecoration(
        color: Theme.of(context).cardTheme.color,
        borderRadius: BorderRadius.circular(16),
      ),
      child: Column(
        children: [
          // Speed indicators
          Row(
            mainAxisAlignment: MainAxisAlignment.spaceEvenly,
            children: [
              _buildSpeedIndicator(
                icon: Icons.arrow_downward,
                label: 'Download',
                speed: stats.downloadSpeed,
                color: Colors.blue,
              ),
              Container(height: 40, width: 1, color:
        Colors.grey.shade300),
              _buildSpeedIndicator(
                icon: Icons.arrow_upward,
                label: 'Upload',
                speed: stats.uploadSpeed,
                color: Colors.green,
              ),
            ],
          ),

          const Divider(height: 24),

          // Data usage and duration
          Row(
            mainAxisAlignment: MainAxisAlignment.spaceEvenly,
            children: [
              _buildInfoItem('Total Down',
          _formatBytes(stats.totalDownload)),
              _buildInfoItem('Total Up',
          _formatBytes(stats.totalUpload)),
              _buildInfoItem('Duration',
          _formatDuration(stats.connectionDuration)),
```

```
            _buildInfoItem('Ping', '${stats.pingMs}ms'),
          ],
        ),

        // Mini speed chart
        if (stats.usageHistory.isNotEmpty)
          SizedBox(
            height: 60,
            child: _buildMiniChart(stats.usageHistory),
          ),
      ],
    ),
  );
}

Widget _buildSpeedIndicator({
  required IconData icon,
  required String label,
  required double speed,
  required Color color,
}) {
  return Column(
    children: [
      Icon(icon, color: color, size: 20),
      const SizedBox(height: 4),
      Text(
        _formatSpeed(speed),
        style: const TextStyle(fontSize: 18, fontWeight:
      FontWeight.bold),
      ),
      Text(label, style: TextStyle(fontSize: 12, color:
      Colors.grey.shade600)),
    ],
  );
}

Widget _buildInfoItem(String label, String value) {
  return Column(
    children: [
      Text(value, style: const TextStyle(fontWeight:
      FontWeight.w600)),
      Text(label, style: TextStyle(fontSize: 11, color:
      Colors.grey.shade600)),
    ],
  );
}

Widget _buildMiniChart(List<DataUsagePoint> history) {
  // Simple sparkline implementation
  final maxSpeed = history.fold<double>(0, (max, p) =>
      p.downloadSpeed > max ? p.downloadSpeed : max);
```

```
    return CustomPaint(
      size: const Size(double.infinity, 60),
      painter: SpeedChartPainter(
        history: history,
        maxSpeed: maxSpeed > 0 ? maxSpeed : 1,
      ),
    );
  }

  String _formatSpeed(double bytesPerSecond) {
    if (bytesPerSecond < 1024) return '$
        {bytesPerSecond.toStringAsFixed(0)} B/s';
    if (bytesPerSecond < 1024 * 1024) return '${(bytesPerSecond /
        1024).toStringAsFixed(1)} KB/s';
    return
        '${(bytesPerSecond / (1024 * 1024)).toStringAsFixed(1)}
        MB/s';
  }

  String _formatBytes(int bytes) {
    if (bytes < 1024) return '$bytes B';
    if (bytes < 1024 * 1024) return '${(bytes /
        1024).toStringAsFixed(1)} KB';
    if (bytes < 1024 * 1024 * 1024) return '${(bytes / (1024 *
        1024)).toStringAsFixed(1)} MB';
    return '${(bytes / (1024 * 1024 * 1024)).toStringAsFixed(2)}
        GB';
  }

  String _formatDuration(int seconds) {
    final duration = Duration(seconds: seconds);
    final hours = duration.inHours;
    final minutes = duration.inMinutes % 60;
    final secs = duration.inSeconds % 60;
    return '${hours.toString().padLeft(2, '0')}:$
        {minutes.toString().padLeft(2, '0')}:$
        {secs.toString().padLeft(2, '0')}';
  }
}
```

6.5 Haptic Feedback

```
// lib/services/haptic_service.dart
import 'package:flutter/services.dart';

class HapticService {
  /// Light impact for UI interactions (button taps, toggles)
  static Future<void> lightImpact() async {
    await HapticFeedback.lightImpact();
  }

  /// Medium impact for significant actions
```

```dart
  static Future<void> mediumImpact() async {
    await HapticFeedback.mediumImpact();
  }

  /// Heavy impact for critical actions (connect/disconnect)
  static Future<void> heavyImpact() async {
    await HapticFeedback.heavyImpact();
  }

  /// Success feedback for completed operations
  static Future<void> success() async {
    await HapticFeedback.heavyImpact();
  }

  /// Error feedback for failures
  static Future<void> error() async {
    await HapticFeedback.vibrate();
  }

  /// Connection state change feedback
  static Future<void> connectionStateChanged(bool connected)
        async {
    if (connected) {
      await HapticFeedback.heavyImpact();
      await Future.delayed(const Duration(milliseconds: 100));
      await HapticFeedback.lightImpact();
    } else {
      await HapticFeedback.mediumImpact();
    }
  }
}
```

6.6 Deep Linking

```dart
// lib/services/deep_link_service.dart
import 'package:go_router/go_router.dart';

class DeepLinkService {
  static final GoRouter router = GoRouter(
    initialLocation: '/',
    routes: [
      GoRoute(
        path: '/',
        builder: (context, state) => const ConnectionScreen(),
      ),
      GoRoute(
        path: '/connect',
        builder: (context, state) {
          // Handle connect deep link
          WidgetsBinding.instance.addPostFrameCallback((_) {
```

```
          context.read(connectionStateProvider.notifier).connect();
            });
            return const ConnectionScreen();
          },
        ),
        GoRoute(
          path: '/disconnect',
          builder: (context, state) {
            WidgetsBinding.instance.addPostFrameCallback((_) {

          context.read(connectionStateProvider.notifier).disconnect();
            });
            return const ConnectionScreen();
          },
        ),
        GoRoute(
          path: '/servers',
          builder: (context, state) => const
          ServerSelectionScreen(),
        ),
        GoRoute(
          path: '/settings',
          builder: (context, state) => const SettingsScreen(),
        ),
        GoRoute(
          path: '/settings/split_tunneling',
          builder: (context, state) => const SplitTunnelingScreen(),
        ),
      ],
    );
}
```

Android manifest deep link configuration:

```
<!-- AndroidManifest.xml -->
<activity
    android:name=".MainActivity"
    android:exported="true"
    android:launchMode="singleTop">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />
        <category
        android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
    <intent-filter>
        <action android:name="android.intent.action.VIEW" />
        <category android:name="android.intent.category.DEFAULT" /
        >
        <category
        android:name="android.intent.category.BROWSABLE" />
        <data android:scheme="helixvpn" />
```

```
    </intent-filter>
</activity>
```

iOS URL scheme configuration:

```
<!-- Runner/Info.plist -->
<key>CFBundleURLTypes</key>
<array>
    <dict>
        <key>CFBundleURLName</key>
        <string>com.helix.vpn</string>
        <key>CFBundleURLSchemes</key>
        <array>
            <string>helixvpn</string>
        </array>
    </dict>
</array>
```

6.7 Share Extension (iOS)

```
// ShareExtension/ShareViewController.swift
import UIKit
import Social

class ShareViewController: SLComposeServiceViewController {

    override func isContentValid() -> Bool {
        return true
    }

    override func didSelectPost() {
        // Handle shared content - e.g., import WireGuard config
        from file
        if let item = extensionContext?.inputItems.first as?
        NSExtensionItem,
           let attachments = item.attachments {

            for attachment in attachments {
                if
        attachment.hasItemConformingToTypeIdentifier("public.data")
        {
                    attachment.loadItem(forTypeIdentifier:
        "public.data") {
                        (data, error) in
                        if let url = data as? URL,
                           let configData = try? Data(contentsOf:
        url) {
                            // Save to App Group for main app to
        pick up
                            self.saveSharedConfig(configData,
        filename: url.lastPathComponent)
                        }
                    }
```

```
                }
            }
        }

        extensionContext?.completeRequest(returningItems: nil)
    }

    private func saveSharedConfig(_ data: Data, filename: String)
        {
        guard let containerURL = FileManager.default
            .containerURL(forSecurityApplicationGroupIdentifier:
        "group.com.helix.vpn") else {
            return
        }

        let configURL =
        containerURL.appendingPathComponent("shared_configs")
        try? FileManager.default.createDirectory(at: configURL,
        withIntermediateDirectories: true)

        let fileURL = configURL.appendingPathComponent(filename)
        try? data.write(to: fileURL)
    }

    override func configurationItems() -> [Any]! {
        return []
    }
}
```

---

7. Build & Distribution

7.1 Android Build Configuration

Gradle Flavors:

```
// android/app/build.gradle
android {
    flavorDimensions += "distribution"

    productFlavors {
        playStore {
            dimension "distribution"
            applicationIdSuffix ".play"
            resValue "string", "app_name", "Helix VPN"
        }
        fdroid {
            dimension "distribution"
            applicationIdSuffix ".fdroid"
            resValue "string", "app_name", "Helix VPN (F-Droid)"
            // No Google Play Services dependencies
```

```
        }
        direct {
            dimension "distribution"
            resValue "string", "app_name", "Helix VPN"
        }
    }
}
```

Build commands:

```
# Google Play Store (AAB)
flutter build appbundle --release --flavor playStore

# F-Droid (APK per ABI)
flutter build apk --release --flavor fdroid --split-per-abi

# Direct distribution (APK)
flutter build apk --release --flavor direct

# Debug build
flutter run --flavor direct --debug
```

7.2 iOS Build Configuration

```
# Build iOS archive for App Store
flutter build ipa --release --export-options-plist=ios/
        ExportOptions.plist

# Build for TestFlight
flutter build ipa --release --export-method=app-store-connect

# Build for development device
flutter run --release

# Build for simulator (VPN functionality won't work)
flutter build ios --simulator
```

ExportOptions.plist:

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN"
  "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
    <key>method</key>
    <string>app-store-connect</string>
    <key>teamID</key>
    <string>YOUR_TEAM_ID</string>
    <key>uploadSymbols</key>
    <true/>
    <key>uploadBitcode</key>
    <false/>
```

```
    <key>signingStyle</key>
    <string>manual</string>
    <key>signingCertificate</key>
    <string>Apple Distribution</string>
    <key>provisioningProfiles</key>
    <dict>
        <key>com.helix.vpn</key>
        <string>Helix VPN App Store</string>
        <key>com.helix.vpn.packet-tunnel</key>
        <string>Helix VPN Extension App Store</string>
    </dict>
</dict>
</plist>
```

7.3 HarmonyOS Build Configuration

```
# Prerequisites
# 1. Install DevEco Studio
# 2. Configure HarmonyOS SDK (API 12+)
# 3. Set up signing certificates from Huawei Developer

# Build HAP package
flutter build hap --release

# Output:
# build/ohos/outputs/default/entry-default-signed.hap

# For debug build
flutter run --debug

# Install on device (requires hdc tool)
hdc install -r build/ohos/outputs/default/entry-default-signed.hap
```

7.4 CI/CD Pipeline for Mobile Builds

```
# .github/workflows/mobile-build.yml
name: Mobile Build Pipeline

on:
  push:
    branches: [main, develop]
  pull_request:
    branches: [main]

jobs:
  build-android:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
```

```yaml
    - name: Setup Flutter
      uses: subosito/flutter-action@v2
      with:
        flutter-version: '3.22.0'

    - name: Setup Java
      uses: actions/setup-java@v4
      with:
        java-version: '17'
        distribution: 'temurin'

    - name: Setup Rust
      uses: dtolnay/rust-action@stable

    - name: Install cargo-ndk
      run: cargo install cargo-ndk

    - name: Add Android Rust targets
      run: |
        rustup target add aarch64-linux-android
        rustup target add armv7-linux-androideabi
        rustup target add x86_64-linux-android

    - name: Build Rust core
      run: |
        cd helix-core
        cargo ndk -t armeabi-v7a -t arm64-v8a -t x86_64 -o ../
      android/app/src/main/jniLibs build --release

    - name: Get Flutter dependencies
      run: flutter pub get

    - name: Build Play Store AAB
      run: flutter build appbundle --release --flavor playStore

    - name: Build F-Droid APKs
      run: flutter build apk --release --flavor fdroid --split-
      per-abi

    - name: Upload artifacts
      uses: actions/upload-artifact@v4
      with:
        name: android-builds
        path: |
          build/app/outputs/bundle/playStoreRelease/
          build/app/outputs/apk/fdroid/release/

build-ios:
  runs-on: macos-latest
  steps:
    - uses: actions/checkout@v4
```

```yaml
      - name: Setup Flutter
        uses: subosito/flutter-action@v2
        with:
          flutter-version: '3.22.0'

      - name: Setup Rust
        uses: dtolnay/rust-action@stable

      - name: Add iOS Rust targets
        run: |
          rustup target add aarch64-apple-ios

      - name: Build Rust core
        run: |
          cd helix-core
          cargo build --target aarch64-apple-ios --release

      - name: Install dependencies
        run: flutter pub get

      - name: Build iOS
        run: flutter build ios --release --no-codesign

      - name: Build IPA
        run: flutter build ipa --release --no-codesign

      - name: Upload artifacts
        uses: actions/upload-artifact@v4
        with:
          name: ios-build
          path: build/ios/ipa/

  build-harmonyos:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Setup Flutter (HarmonyOS)
        uses: subosito/flutter-action@v2
        with:
          flutter-version: '3.22.0-ohos'
          channel: 'stable'

      - name: Setup HarmonyOS SDK
        run: |
          # Download and configure HarmonyOS SDK
          wget https://contentcenter-drcn.dbankcdn.com/harmonyos/
        sdk/HarmonyOS-SDK-API12.tar.gz
          tar -xzf HarmonyOS-SDK-API12.tar.gz -C $HOME/
          echo "OHOS_SDK=$HOME/HarmonyOS-SDK" >> $GITHUB_ENV

      - name: Setup Rust
```

```
        uses: dtolnay/rust-action@stable

      - name: Build Rust core for HarmonyOS
        run: |
          cd helix-core
          cargo build --target aarch64-linux-ohos --release

      - name: Build HAP
        run: flutter build hap --release

      - name: Upload artifacts
        uses: actions/upload-artifact@v4
        with:
          name: harmonyos-build
          path: build/ohos/outputs/
```

7.5 Code Signing and Certificates

Android:

Certificate Type	Purpose	Location
Debug keystore	Development builds	`~/.android/debug.keystore`
Release keystore	Play Store / distribution	Secure CI secret
Upload key	Google Play signing	Managed by Google Play

```
# Generate release keystore
keytool -genkey -v \
  -keystore helix-release.keystore \
  -alias helix \
  -keyalg RSA \
  -keysize 2048 \
  -validity 10000

# Extract certificate fingerprint for API key registration
keytool -list -v -keystore helix-release.keystore -alias helix
```

iOS:

Certificate	Purpose	Type
Apple Development	Development builds	Development
Apple Distribution	App Store / TestFlight	Distribution
Network Extension entitlement	VPN capability	Self-serve

```
# Certificates managed through Apple Developer portal:
# https://developer.apple.com/account/resources/certificates/list

# Provisioning profiles:
# 1. Helix VPN App Development (development cert)
# 2. Helix VPN App Store (distribution cert)
# 3. Helix VPN Extension Development
# 4. Helix VPN Extension App Store
```

HarmonyOS:

Certificate	Source
App signing certificate	Huawei AppGallery Connect
App signing private key (.p12)	Generated in DevEco Studio
Profile file (.p7b)	AppGallery Connect

```
# Generate signing materials in DevEco Studio:
# Build -> Generate Key and CSR
# Upload CSR to AppGallery Connect
# Download certificate and profile
# Configure signing in build-profile.json5
```

7.6 Distribution Channels Summary

Platform	Channel	Format	Requirements
Android	Google Play	AAB	Developer account ($25 one-time), privacy policy
Android	F-Droid	APK	FOSS license, reproducible builds, F-Droid metadata
Android	Direct/APK	APK	Self-hosted, user enables "Unknown sources"
iOS	App Store	IPA	Developer Program ($99/year), app review
iOS	TestFlight	IPA	App Store Connect, up to 10,000 external testers
HarmonyOS	AppGallery	HAP	Huawei Developer account, security review

---

Appendix A: File Structure

```
helix-vpn-mobile/
├── android/                          # Android platform code
│   ├── app/
│   │   ├── build.gradle
```

```
│   │   ├── proguard-rules.pro
│   │   └── src/
│   │       ├── main/
│   │       │   ├── AndroidManifest.xml
│   │       │   ├── kotlin/
│   │       │   │   └── com/helix/vpn/
│   │       │   │       ├── MainActivity.kt
│   │       │   │       ├── vpn/
│   │       │   │       │   ├── HelixVpnService.kt
│   │       │   │       │   ├── VpnNotificationHelper.kt
│   │       │   │       │   └── HelixVpnTileService.kt
│   │       │   │       ├── model/
│   │       │   │       │   ├── VpnConfig.kt
│   │       │   │       │   └── ConnectionStats.kt
│   │       │   │       └── rust/
│   │       │   │           └── HelixRustBridge.kt
│   │       │   └── cpp/           # CMake NDK bridge (if needed)
│   │       └── playStore/
│   │       └── fdroid/
│   └── build.gradle
│
├── ios/                              # iOS platform code
│   ├── Runner/
│   │   ├── AppDelegate.swift
│   │   ├── Info.plist
│   │   └── Runner.entitlements
│   ├── PacketTunnelProvider/
│   │   ├── PacketTunnelProvider.swift
│   │   ├── Info.plist
│   │   └── PacketTunnelProvider.entitlements
│   ├── HelixVpnWidget/
│   │   └── HelixVpnWidget.swift
│   ├── ShareExtension/
│   │   └── ShareViewController.swift
│   └── Runner.xcodeproj/
│
├── ohos/                             # HarmonyOS platform code
│   ├── entry/
│   │   ├── src/main/
│   │   │   ├── ets/
│   │   │   │   ├── entryability/
│   │   │   │   │   └── EntryAbility.ets
│   │   │   │   ├── vpnability/
│   │   │   │   │   └── HelixVpnAbility.ets
│   │   │   │   ├── rust/
│   │   │   │   │   └── HelixRustBridge.ets
│   │   │   │   └── widget/
│   │   │   │       └── HelixWidgetAbility.ets
│   │   │   └── cpp/
│   │   │       ├── napi_helix_bridge.cpp
│   │   │       └── CMakeLists.txt
│   │   └── module.json5
```

```
│   └── build-profile.json5
│
├── lib/                              # Flutter Dart code
│   ├── main.dart
│   ├── app.dart
│   ├── theme/
│   │   └── helix_theme.dart
│   ├── models/
│   │   ├── connection_stats.dart
│   │   ├── server.dart
│   │   └── app_settings.dart
│   ├── providers/
│   │   ├── connection_provider.dart
│   │   ├── server_provider.dart
│   │   └── settings_provider.dart
│   ├── services/
│   │   ├── vpn_service.dart
│   │   ├── biometric_service.dart
│   │   ├── haptic_service.dart
│   │   ├── notification_service.dart
│   │   ├── auto_connect_service.dart
│   │   └── deep_link_service.dart
│   ├── ui/
│   │   ├── screens/
│   │   │   ├── connection_screen.dart
│   │   │   ├── server_selection_screen.dart
│   │   │   ├── settings_screen.dart
│   │   │   ├── split_tunneling_screen.dart
│   │   │   └── account_screen.dart
│   │   ├── widgets/
│   │   │   ├── connection_button.dart
│   │   │   ├── connection_stats_panel.dart
│   │   │   ├── country_flag.dart
│   │   │   ├── server_list_tile.dart
│   │   │   └── speed_chart.dart
│   │   └── platform/
│   │       └── platform_adapters.dart
│   └── generated/                    # flutter_rust_bridge
generated
│
├── helix-core/                       # Shared Rust core
│   ├── src/
│   │   ├── lib.rs
│   │   ├── android/
│   │   ├── ios/
│   │   ├── harmonyos/
│   │   ├── protocol/
│   │   ├── crypto/
│   │   └── state/
│   └── Cargo.toml
│
├── pubspec.yaml
```

```
├── build.gradle
└── README.md
```

Appendix B: Platform Comparison Matrix

Feature	Android	iOS	HarmonyOS
VPN API	VpnService	NEPacketTunnelProvider	VpnExtensionAbilit
Min OS	API 26 (8.0)	iOS 15	API 12 (NEXT)
Code Reuse	85-90%	85-90%	80-85%
Background	Foreground service	System extension	ExtensionAbility
Memory Limit	~200MB+	~15MB	Unknown
Split Tunneling	App + Route-based	Route-based only	App-based
Always-On VPN	Yes (system)	On-Demand rules	Limited
Kill Switch	Lockdown mode	Limited	isBlocking flag
Quick Toggle	QS Tile	Control Center	Form card
Biometrics	BiometricPrompt	Face ID / Touch ID	System biometric
Credential Store	Keystore	Keychain	HUKS
Notifications	FCM	APNs	HMS Push
Distribution	Play/F-Droid/ Direct	App Store/TestFlight	AppGallery

---

This specification is a living document. As the MVP2 project evolves, implementation details may be refined based on prototyping results and platform API changes.

End of Document